



Multi-object tracking review: retrospective and emerging trend

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Abstract

Multi-object tracking (MOT) is a critical task involving detecting and continuously tracking multiple objects within a video sequence. It is widely used in various fields, such as autonomous driving and intelligent security. In recent years, deep learning architectures have effectively promoted the development of MOT. However, this task poses significant challenges regarding accuracy due to occlusion/truncation, light variation, camera movement. Researchers have proposed many methods to address these issues to reduce trajectory fragmentation, identity switches, and missing targets. To better understand these advancements, it is essential to categorize the approaches based on their methodologies. This article reviewed the recent development of MOT, divided into Tracking by Detection (TBD) and End-to-End (E2E). By introducing and comparing the two types of tracking algorithms, readers can quickly understand the current development status of MOT. Meanwhile, this review summarizes the links to open-source code of excellent algorithms and common benchmark datasets in the appendix. And provide a unified MOT toolkit that includes evaluation and visualization at <https://github.com/guanzhiyu817/MOT-tools>. In addition, this review discusses the future directions of MOT, specifically cross-modal reasoning.

Keywords Multi-object tracking · Tracking by detection · End-to-end · Object detection · Cross-modal

1 Introduction

Multi-object Tracking (MOT) is a research hotspot in computer vision, which focuses on locating and tracking multiple moving objects in the video sequence. It has a wide range of applications, including autonomous driving, intelligent security, robot navigation and medical imaging. With the success of deep learning in detection tasks (Ahmed et al. 2023), MOT based on deep learning has gradually become the fundamental paradigm in the past

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decade. Most existing methods take various videos as input, as shown in Fig. 1, typically tracking pedestrian or vehicle objects to assist in achieving structured analysis. It is inevitable to encounter occlusion of objects, changes in video lighting, and camera movement (Jiao et al. 2023). These cases often lead to detection failures or incorrect associations. To deal with these problems, many excellent methods have been proposed (Ciaparrone et al. 2020; Pal et al. 2021; Luo et al. 2021; Rakai et al. 2022), one after another. According to whether detection and association are trained together, these approaches (Tang et al. 2017; Bergmann et al. 2019; Zhang et al. 2019; Guo et al. 2021) are divided into two categories in the paper: Tracking by Detection (TBD) and End-to-End (E2E).

In TBD, detection and association are independent of each other. Various successful detectors are employed to locate objects in consecutive frames, such as Faster R-CNN (Ren et al. 2015), DPM (Felzenszwalb et al. 2009), SDP (Yang et al. 2016), and so on. And then, different similarity measures (Qin et al. 2023; Leal-Taixé et al. 2016) and matching strategies (Liu et al. 2004; Henriques et al. 2011) based on localization, motion, and appearance are exploited to achieve trajectory association. For some offline methods (Pang et al. 2020), there are also many post-processing operations based on the time interval and position spacing to optimize trajectory integrity. Contrary to TBD, the detection and association of E2E are designed in a unified framework for joint training (Zhang et al. 2019; Guo et al. 2021). Representative works of E2E typically use convolutional or Transformer blocks as feature encoders (Zhang et al. 2021; You et al. 2023; Meinhardt et al. 2022; Sun et al. 2020). In CNN-based frameworks (Zhang et al. 2021; You et al. 2023), some RNN or LSTM modules are often introduced to enhance temporal features for location and tracking. For Transformer networks (Meinhardt et al. 2022; Sun et al. 2020), selecting features and attention design often play an important role in MOT. As the designs of TBD MOT (Mandel et al. 2023; Cao et al. 2023) and E2E MOT (Li et al. 2023b; Zhao et al. 2023b) have matured, the tracking community has predominantly focused on these two paradigms in recent years, resulting in significant enhancements in tracking performance across MOT datasets.

In order to help readers quickly understand the current development status of MOT, this article starts from the challenges of tracking and comprehensively elaborates and compares the existing methods of TBD and E2E. The primary challenge of MOT lies in achieving distinctive tracking features to adapt the different factors (i.e., occlusion, illumination, camera) in the real world. Taking handling occlusion/truncation as an example, TBD methods focus



Fig. 1 Visualization explanation of MOT. **a** depicts the trajectory generated in pedestrian multi-target tracking, where the object with transparency represents the initial position of the pedestrian, and the object without transparency represents the current position. **b** illustrates the trajectory in vehicle multi-target tracking, where the vehicle with transparency denotes its position in the previous frame, and the object without transparency denotes the current position of the vehicle

on appearance learning to confront challenges (Mandel et al. 2023; Zhang et al. 2022b), while E2E methods strengthen temporal clues to infer trajectory blanks (Zhang et al. 2021; You et al. 2023). For camera movement, TBD may utilize global and local features to adapt to environmental changes (Qin et al. 2023; Cao et al. 2023), while E2E can obtain the most suitable tracking features for the current scene by selecting tokens (Meinhardt et al. 2022; Sun et al. 2020). Previous review papers have primarily focused on discussing problems and methodologies, requiring readers to independently search for open-source code to further their understanding. To be reader-friendly, the open-source codes of these methods are collected and organized in this paper, which provides a comprehensive analysis and discussion from theory to implementation.

In addition, this paper also quantitatively and qualitatively compared outstanding representative works. The advantages and disadvantages of different methods were demonstrated in terms of accuracy and efficiency so that readers could choose baselines according to actual needs. On this basis, a toolkit is provided for evaluation and visualization, which is friendly to beginners and beneficial for promoting initial research of MOT. This review also meticulously reviewed the prevailing evaluation metrics applied to datasets relevant to the past several years. Rigorous analysis of diverse MOT datasets and associated evaluation metrics is a foundation for forecasting emerging domains warranting further exploration. The subsequent sections of this review elaborate on these perspectives, providing a comprehensive understanding of contemporary trends and unexplored domains in MOT research.

The contributions of this work lie three-fold:

- 1) It classifies TBD methods into Object Detection-centric and Data Association-centric approaches, and E2E methods into CNN-based and Transformer-based approaches, offering readers intuitive insights;
- 2) It organizes extensive open-source code and download links for prevalent MOT datasets, and provides a toolkit to evaluate and visualize experimental results, enabling newcomers to easily and quickly identify the limitations of algorithms in handling challenges, thereby fostering the development of MOT;
- 3) It predicts and outlines the future directions and emerging trends in MOT research from four perspectives, particularly its integration with fields like healthcare and AI agents, which are expected to drive further advancements.

2 Challenges of MOT

Many challenges emerge within the realm of MOT, encompassing a spectrum of intricacies akin to those encountered in single-object tracking. These challenges encapsulate occlusion, truncation, deformation, motion blur, crowded scenes, light variations and camera movement. Additionally, multi-object tracking brings forth distinctive complexities that extend beyond the challenges of single-object tracking. These complexities include identification switches and the potential for interference between similar objects, further complicating the tracking process.

The diverse application scenarios where MOT finds utility, including autonomous vehicle driving, video surveillance, robot navigation, and behavior recognition, each present distinct challenges. Despite significant strides in research on MOT algorithms in recent

years, obstacles persist due to object appearances and non-linear motion dynamics alterations. Consequently, accurate MOT remains a challenging domain within computer vision. The challenges of MOT are shown in Fig. 2.

In Fig. 2, as shown in (a), when the target pedestrian 1 is occluded by other pedestrians 2, the phenomenon of trajectory fragmentation is evident. As depicted in (b), due to the change in the external lighting environment, the appearance of target pedestrian 1 changed, resulting in the occurrence of the identification switch (ID switch) phenomenon during the tracking process. As illustrated in (c), during the movement of target vehicle 1, factors such as turning may cause the camera to shift, potentially leading to target loss.

2.1 Occlusion/truncation

Challenges posed by occlusion and truncation are pivotal and recurrent hurdles within the MOT domain. Occlusion occurs when other objects or obstacles partially or completely obscure an object, preventing the tracking system from directly observing the obscured object (Specker et al. 2021). Illustrated in Fig. 2a, When a blue target pedestrian encounters a black pedestrian, the blue pedestrian becomes partially occluded by the black one. This partial occlusion reduces the visibility of the target pedestrian, presenting challenges to MOT. The challenges may lead to situations such as trajectory fragmentation. In such cases, maintaining accurate tracking becomes challenging.

It is still a complex problem to successfully solve the tracking misalignment caused by occlusion and complex tracking scenes (Zhu et al. 2018). Addressing occlusion challenges involves employing various strategies. Integrating multimodal fusion (Lu et al. 2022), combining diverse sensor information from modalities like depth cameras or lidar, proves crucial in enhancing the understanding of the spatial disposition of the object, even amidst occluded

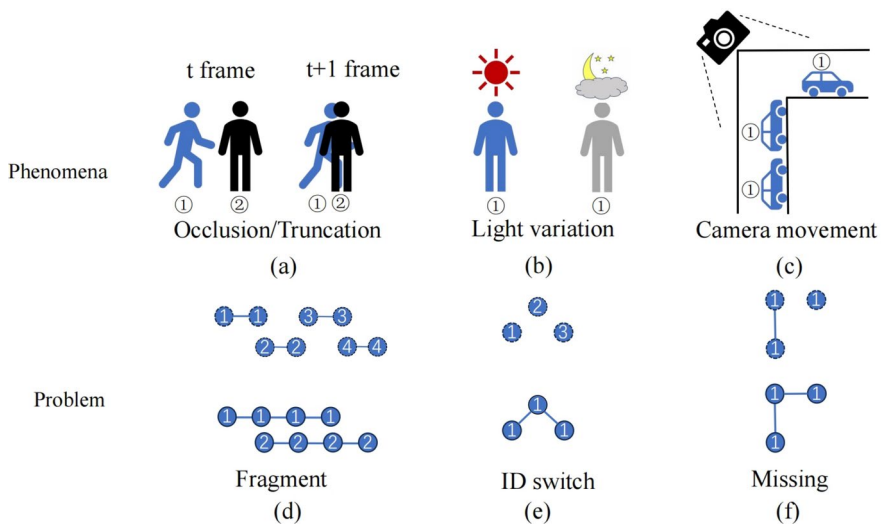


Fig. 2 Challenges of MOT. Figures a, b, and c illustrate objective phenomena in MOT: occlusion/truncation, light variation, and camera movement, respectively. d, e, and f showcase the problems and challenges in MOT, namely fragment, ID switch, and missing, respectively. The dashed circle represents the predicted result, while the solid circle represents the ground truth

scenarios. Additionally, leveraging the appearance characteristics of objects to model their visual attributes enables the identification and association of the object, even when occlusion or truncation affects its complete visibility (Kim et al. 2021). Additionally, Most errors in occlusion scenarios stem from motion models relying on linear motion assumptions. Tokmakov et al. (2021) proposed the PermaTrack model, which operates under the assumption that any physical object existing at a given moment is likely to persist throughout the entire tracking period. This assumption proves beneficial in predicting the location of objects, even when they are fully occluded during specific intervals. Cao et al. (2023) proposed OC-SORT that employs target observations, precisely measurements obtained from the target detector, to compute virtual trajectories. This strategy is designed to mitigate the accumulation of errors in the filter parameters, a common challenge faced during occlusion periods.

These methodological interventions significantly enhance the efficacy of trackers in managing object occlusion or truncation scenarios. Moreover, the challenge of occlusion remains a dynamically evolving research area within MOT, driving ongoing advancements in methodologies and techniques. The continual development of novel strategies aims to fortify the resilience and adaptability of trackers in confronting occlusion-related challenges, emphasizing the sustained effort to enhance the robustness of trackers in occlusion-laden scenarios.

2.2 Light variation

The challenge of illumination variation is critical in MOT, representing recurrent fluctuations in ambient lighting conditions that significantly alter the appearance of objects across disparate video frames. As illustrated in Fig. 2b, pronounced alterations in outdoor lighting induce conspicuous changes in the characters depicted within the outlined box. Transitions from indoor to outdoor environments particularly trigger notable lighting modifications, leading to shifts in the color palette, texture, and other visual attributes of the tracked objects (Wang et al. 2023c). Such variability in appearance results in the loss of critical visual cues necessary for accurate object tracking, leading to potential targets missing.

Adaptive methodologies have been employed to alleviate the adversities stemming from illumination variations. Techniques such as appearance consistency detection (Sun et al. 2022) and integrating illumination-invariant features (Bouraya Jr and Belangour 2021) have emerged to represent objects using features robust to illumination fluctuations, like gradient direction histograms. These endeavors aim to mitigate the influence of illumination variations on object identification, thereby enhancing the resilience of the tracking system against lighting fluctuations. In addition, in multimodal representation fusion (Cai et al. 2023), introducing supplementary hyperparameters has led to divergent performances across varying lighting and weather conditions. This disparity hinders the optimal balancing of the contribution of each input domain to the overall cost. Addressing this challenge, Mancusi et al. (2023) proposed the TrackFlow model, synthesizing multiple input costs and displacements into a unified output indicator, offering a comprehensive perspective to subsequent allocation optimization predicaments.

Investigating light variations in the context of MOT remains an area that requires further exploration. Understanding the impact of lighting changes on MOT systems is crucial for improving their robustness and performance in real-world scenarios. Particularly in intricate outdoor settings or environments characterized by erratic lighting dynamics, strategies to

cope with illumination fluctuations emerge as a pivotal research trajectory within the multifaceted domain of MOT.

2.3 Camera movement

As surveillance video technology evolves, camera movement and MOT convergence assume paramount significance, enabling automated monitoring across vast observation areas. Object matching the camera movement forms the crux, aiming to establish correspondences among objects scattered across distinct image sequences. The outcomes of these matches significantly influence subsequent data fusion endeavors (Dockstader and Tekalp 2001), thereby impacting the performance of the MOT system. The domain of MOT grapples with numerous challenges, notably highlighted in Fig. 2c, where variations in camera perspectives result in discrepancies in object scales and configurations across different camera views.

Addressing the challenges stemming from varying camera positions necessitates the adoption of attitude estimation methodologies (Schmauser et al. 2022). These technologies facilitate object position and orientation estimation under diverse viewing angles, mitigating the adverse impact of positional discrepancies. Moreover, the arrangement and positioning of cameras play a tangible role in the efficacy of MOT. Diverse camera layouts can cause disparities in tracking coverage and observation quality, prompting optimizing layouts to enhance tracking performance (Zou et al. 2018).

Resolving these intricate challenges calls for deploying cutting-edge technologies, such as multi-sensor fusion strategies (Ouyang et al. 2022; Senel et al. 2023) and advanced deep learning methodologies (Ciaparrone et al. 2020; You and Jiang 2020). These approaches offer an integrative mechanism to harmonize data from various sensors, including radar, infrared cameras, and drones, into a cohesive data stream. Additionally, Zhang et al. (2022a) proposed MUTR3D, which introduces 3D trajectory queries to effectively model spatially and appearance-coherent trajectories for objects that manifest across multiple cameras and frames. Simultaneously, integrating camera transformations facilitates the alignment of the 3D tracker with its corresponding observations in 2D images. Pang et al. (2023) PF-Track framework stands out as an end-to-end paradigm. Leveraging spatio-temporal cues, this framework utilizes attention mechanisms to extract object dynamics and interactions, subsequently utilizing this information for trajectory refinement and long-term trajectory predictions.

Investigating camera position variation is an up-and-coming research area within MOT. Notably, a limited body of research findings exists regarding vision-based 3D object tracking, and the domain of 3D vision trackers holds multifaceted potential for applications across diverse domains.

3 Related work

Multi-object tracking (Hassan et al. 2024) involves object detection, feature extraction, and data association. Initially, object detection occurs in each frame of images or videos. Subsequently, features like color and texture are extracted from the detected objects. Simultaneously, the historical state of the object proves to be crucial, encompassing key attributes

such as the position and velocity of the object, which helps the tracking algorithm predict the future position of the object. Ultimately, data association addresses one of the fundamental challenges of multi-object tracking by establishing connections between objects in the current frame and data from previous or multiple frames, ensuring the continuity of object identity (Son et al. 2017).

Recent multi-object tracking techniques primarily fall into TBD and E2E multi-object tracking, illustrated in Fig. 3. This figure delineates various TBD and E2E MOT methods commonly employed in the field. These classifications delineate the core methodologies employed in these two prevalent paradigms of MOT.

3.1 Tracking by detection

3.1.1 Object detection-centric approaches in MOT

Within the TBD MOT paradigm (Cao et al. 2023; Mandel et al. 2023; Zhang et al. 2022b; Chen et al. 2017; Bochinski et al. 2017; Qin et al. 2023; Cetintas et al. 2023; He et al.

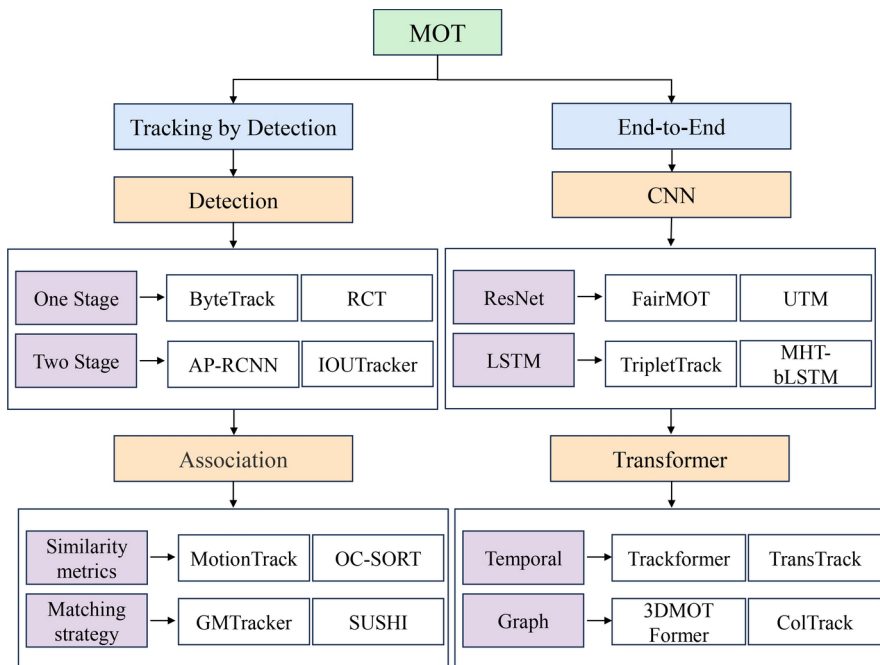


Fig. 3 Tracking by Detection and End-to-End MOT. In TBD MOT, the classification primarily revolves around Object Detection-centric and Data Association-centric approaches in MOT. The landscape of Object Detection-centric approaches reveals that ByteTrack (Zhang et al. 2022c), RCT (Mandel et al. 2023), AP-RCNN (Chen et al. 2017), IOUTracker (Bochinski et al. 2017). The landscape of Data Association-centric approaches reveals that MotionTrack (Qin et al. 2023), OC-SORT (Cao et al. 2023), GMTracker (He et al. 2021a), SUSHI (Cetintas et al. 2023). In E2E MOT, the categorizations include CNN-based MOT and Transformer-based MOT. The landscape of CNN-based MOT reveals that FairMOT (Zhang et al. 2021), UTM (You et al. 2023), TripletTrack (Marinello et al. 2022), MHT-bLSTM (Kim et al. 2018). The landscape of Transformer-based MOT reveals that Trackformer (Meinhardt et al. 2022), TransTrack (Sun et al. 2020), 3DMOTFormer (Ding et al. 2023), ColTrack (Liu et al. 2023b)

2021a), models within this framework rely on precise detections to identify objects. Subsequently, these models establish connections between objects in a distinct phase using appearance features and historical information. It is noteworthy that Object Detection-centric approaches (Zhang et al. 2022b; Mandel et al. 2023; Chen et al. 2017; Bochinski et al. 2017; Kim et al. 2015) within this framework emphasize the initial stage, namely the object detection part.

Traditional methods for object detection encompass algorithms such as Scale-Invariant Feature Transform (SIFT) (Ng and Henikoff 2003), Histogram of Oriented Gradient (HOG) (Dalal and Triggs 2005), and Deformable Part Model (DPM) (Felzenszwalb et al. 2008). SIFT is a classical image feature extraction algorithm that identifies stable feature points across scales and rotations, making it robust against changes in viewpoint and partial occlusions. Its key stages involve keypoint detection, localization, description, and matching. Moreover, association detection is achieved through temporally sparse (Leal-Taixé et al. 2011; Han et al. 2022) or dense graph optimization (Zhang et al. 2015; Keuper et al. 2018). Unlike CNNs, SIFT relies on handcrafted features and does not adapt to new data without retraining. It is also computationally intensive due to the feature extraction process. While HOG is invariant to geometric and optical deformations and is effective for rigid objects. It involves computing histograms of gradient directions in localized regions (cells) and normalizing them within blocks to enhance robustness against illumination changes. However, HOG features are monotonous and do not capture the spatial hierarchies present in images as CNNs extract through their layered structure. Finally, DPM is a part-based detection algorithm that improves upon HOG by accounting for deformation in objects. It decomposes an object into multiple parts, each of which can deform, enhancing detection flexibility. But DPM lacks the ability to generalize across different scenes. So, due to the dependence on handcrafted features and lack of generalization capabilities, traditional methods like SIFT, HOG, and DPM gradually replaced by CNNs.

In recent years, deep learning-based methods have become the mainstream approach for a multitude of computer vision tasks. Deep learning-based object detection algorithms leverage Convolutional Neural Networks (CNNs) (Gu et al. 2018) to autonomously learn features essential for object detection and classification (Ahmed et al. 2023). CNNs transform the original input data into more abstract, higher-dimensional features, possessing robust expression capabilities and generalization across complex scenes. This characteristic enables them to adeptly fulfill diverse industrial application requirements, such as utilizing multimodal medical imaging for cancer detection (Karol et al. 2024) and detecting occluded items in X-ray baggage inspections (Akcaay and Breckon 2022).

Deep learning-based object detection algorithms fall primarily into two categories: one- and two-stage object detection. The two-stage detection involves generating candidate regions and then utilizing CNNs for classification and regression correction. Two-stage detection is more suitable for scenarios with high requirements for detection accuracy, such as fine object recognition, medical image analysis, etc. In these scenarios, two-stage detection can provide high-quality object detection results, especially suitable for complex scenes and applications that require high precision. Popular algorithms include R-CNN (Girshick et al. 2014), SPPNet (Purkait et al. 2017), Fast R-CNN (Girshick 2015), Faster R-CNN (Ren et al. 2015), and R-FCN (Dai et al. 2016) etc. These algorithms generally deliver more precise detection results but are relatively slower. In contrast, one-stage detection algorithms do not generate candidate frames and directly extract network features for object classification

and localization. So they have faster inference speed, suitable for real-time applications or scenarios with high speed requirements, such as real-time video surveillance, autonomous driving and smart security. In these scenarios, one-stage detection can quickly identify and locate objects, meeting real-time requirements. Commonly used algorithms in this category are SSD (Liu et al. 2016a), YOLO series (Redmon et al. 2016; Redmon and Farhadi 2017, 2018; Bochkovskiy et al. 2020; Ge et al. 2021), Retinanet (Lin et al. 2017) and so on, all of which are known for their speed advantage.

As depicted in Fig 3, the landscape of TBD MOT reveals that ByteTrack (Zhang et al. 2022b), RCT (Mandel et al. 2023), AP-RCNN (Chen et al. 2017) and IOUTracker (Bochinski et al. 2017) fall under Object Detection-centric approaches. Specifically, AP-RCNN (Chen et al. 2017) and IOUTracker (Bochinski et al. 2017) leverage two-stage detectors, Faster R-CNN (Girshick 2015) and R-CNN (Girshick et al. 2014), respectively. Conversely, ByteTrack (Zhang et al. 2022b) and RCT (Mandel et al. 2023) opt for one-stage detectors, utilizing YOLOX and YOLOv4 (Bochkovskiy et al. 2020), respectively. This classification highlights the distinctions in detector architectures employed by these tracking methods.

Object Detection-centric approaches, due to their simple structure, often achieve quite good inference speed and relatively satisfactory inference accuracy, making them suitable for devices with limited computing power and scenarios with high real-time requirements. However, for scenarios that pursue high precision, this method still falls short, cause some approaches solely considers the final output of the detector, often neglecting the intermediate CNN features. These approaches primarily concentrate on data association strategies, i.e., linking detections to tracklets. However, they exhibit two drawbacks: a lack of discriminative ability to handle occlusions between interacting targets and susceptibility to detection failures. A significant contribution to the Object Detection-centric field is Chen et al. (2017) proposed AP-RCNN framework, which addresses these issues by extracting features from CNN-based detectors to formulate an appearance model. In general, when incorporating appearance features into multi-object tracking tasks, a separate network is typically dedicated to extracting appearance features, which incurs significant additional computational costs. As a result, while tracking accuracy is improved, inference speed is sacrificed. In contrast, AP-RCNN reuses the appearance features from the detection phase, avoiding extra computations while utilizing feature matching, achieving a reasonable balance between speed and accuracy. This approach expands the thinking for subsequent tracking methods, allowing object detection-centric tracking methods to no longer be limited to detection. Different layers can extract various features by utilizing multiple convolutional layers in the detector, thereby refining the tracking effect. The framework of AP-RCNN (Chen et al. 2017) is illustrated in Fig. 4.

As shown in Fig. 4, features are extracted from multiple convolutional layers utilizing the Faster-RCNN (Girshick 2015) detector to create two classifiers: a category classifier and an instance classifier. Moreover, each tracked object's appearance is initially preserved and updated during online tracking. Upon the arrival of a new frame, the historical appearance and the new feature vectors extracted from each candidate region are concatenated to calculate the probability of the same object. The two classifiers share the convolutional layer with the detector, offering the primary advantage that detection and online tracking can be efficiently executed using the convolutional features computed once per image.

In prior research, Kim et al. (2015) proposed MHT-DAM, which leverages advancements in modern detection tracking and the effective development of object appearance

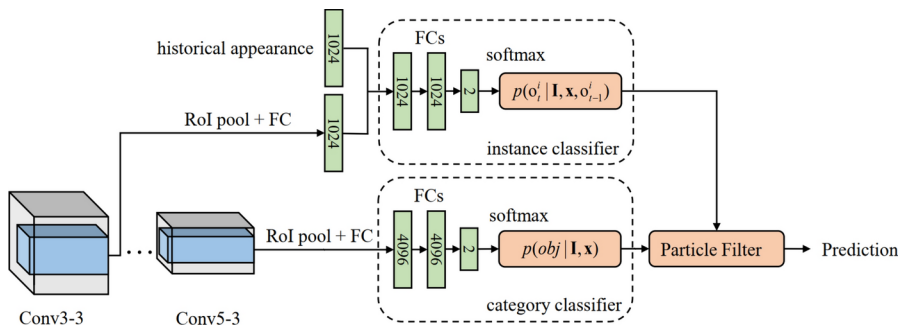


Fig. 4 Framework of AP-RCNN (Chen et al. 2017)

feature representations. This creates new opportunities for Multiple Hypothesis Tracking approaches to enhance tracking accuracy, mainly utilizing contemporary precision object detectors. Yu et al. (2016) proposed that POI is a high-performance detection and deep learning MOT model centered on appearance features. Bochinski et al. (2017) proposed IOUTracker, demonstrates its potential through experimental evaluations using diverse two-stage object detectors, achieving a speed of 100K fps with R-CNN detectors (Girshick et al. 2014). In 2022, Zhang et al. (2022b) proposed ByteTrack, which relies on the high-performance detector YOLOX (Ge et al. 2021) to obtain detection results. ByteTrack employs Kalman filtering, similar to SORT (Bewley et al. 2016), which predicts the trajectory position of the current frame in the subsequent frame for data association. In 2023, Mandel et al. (2023) proposed Robust Confidence Tracking (RCT), an offline MOT algorithm tailored for environments with suboptimal detection quality. RCT relies on precise detection confidence values to enhance tracking quality across the entire pipeline. In 2024, Luo et al. (2024) proposed a simple yet robust framework that formulates object detection as a consistent denoising diffusion process, transforming paired noisy bounding boxes into paired ground truth boxes. This novel progressive denoising diffusion strategy significantly enhances the effectiveness of the tracker, enabling it to distinguish between various objects more accurately. Liu et al. (2024b) introduced a novel MOT method named Pro2Diff, which employs a generative approach. In the forward process, Pro2Diff generates numerous noisy proposals for the tracked image sequence. It then learns the discrepancies between these noisy proposals and the actual bounding boxes of the tracked objects. By progressively refining these noisy proposals, Pro2Diff reconstructs the initial sequence of the actual tracked objects.

3.1.2 Data association-centric approaches in MOT

Within the domain of Data Association-centric approaches (Cao et al. 2023; Zhang et al. 2022b; He et al. 2021a; Yoon et al. 2016) in TBD MOT, models emphasize the data association stage. In this context, data association involves establishing connections between targets identified in previous frames and those in the current frame. Moreover, data association approaches can be broadly categorized into two key phases: similarity calculation (Ristani and Tomasi 2018; Chen et al. 2018; Chu and Ling 2019) and matching strategy (Henriques et al. 2011; Takala and Pietikainen 2007; Fouzia et al. 2019; He et al. 2021a).

The similarity is initially calculated through various methods, including the use of a twin neural network (Leal-Taixé et al. 2016), learned re-identification (ReID) features (Ristani and Tomasi 2018; Pang et al. 2021), selection of detection candidates (Chen et al. 2018), or estimation of affinity (Chu and Ling 2019). Location, motion, and appearance are three important cues to computing the similarity between detections and tracks. Appearance modeling involves abstractly describing the appearance of an object through feature extraction. This extraction phase encompasses single or combined features such as color (Pérez et al. 2002; Possegger et al. 2015), texture, gray-scale, geometric features, or multiple feature transformations (Li and Zhu 2015; Bertinetto et al. 2016) to flexibly model object appearances for various tracking targets in different scenarios. Then, the appearance model assesses the similarity between the current search region and the target to calculate a confidence score, determining the likelihood that the candidate region is indeed the target. This strategy aims to identify and locate the target effectively. Data association methods based on motion trajectory commonly include the Kalman filter (Liu et al. 2004) and particle filter (Jin and Mokhtarian 2007). Object tracking establishes possible positional relationships between frames based on the current object location in the data association phase. This process generates potential candidate positions for the object in the subsequent frame by constructing motion models (Leal-Taixé et al. 2011; Alahi et al. 2016; Robicquet et al. 2016) and defining constraints (Andriyenko and Schindler 2011; Choi and Savarese 2010).

Kalman filter-based methods (Cao et al. 2023; Zhang et al. 2022b; Chiu et al. 2021; Guo and Zhao 2022) for MOT excel in scenarios with assumed linear target movement and brief occlusion periods. Kalman filter can predict the location of targets in future frames, which is crucial for predicting target motion, cause it restricts the previously non-directional predictions and associations in tracking methods to a specific direction, significantly reducing the error rate of prediction and matching. And even when targets are occluded, the Kalman filter can rely on their previous motion states to predict their current locations, helping to alleviate occlusion issues. The most important point is that the Kalman filter, which provides so many effective strategies for multi-object tracking methods, only brings a very small computational overhead and does not slow down the inference speed of the methods. Despite many other new motion modeling strategies being proposed later, the Kalman filter remains a very elegant one among them. However, Linear estimates become notably inaccurate under such circumstances in cases involving nonlinear motion and extended durations. Moreover, in the absence of measurements for updating Kalman filter parameters, the conventional approach involves performing a posteriori updates based on a priori state estimates. Unfortunately, this practice results in error accumulation during occlusion periods, leading to significant deviations in the motion direction. The limitations of the Kalman filter stem from the conflict between theory and reality. The design of the Kalman filter is very sophisticated, and unlike deep learning methods, it is interpretable and therefore controllable. It attempts to approximate the motion trajectory of an object at every moment using a linear function, which is valid when the time interval is very short or the target is moving at a constant speed in a straight line. However, reality often involves many random and complex motions, limitations in video frame rates and computational capability also prevent high-frequency sampling and prediction of the tracked targets. That's why Kalman filter shows its inadequacies. The Kalman filter is an excellent solution for motion modeling in scenarios with limited computational capability, but it is also limited by the limitations in computational capability. Cao et al. (2023) proposed observation-centric SORT (OC-

SORT), a Data Association-centric approach to address this limitation. OC-SORT leverages object observations-provided by an object detector-to compute virtual trajectories during occlusion periods. This approach mitigates the error accumulation of filter parameters, allowing for more effective correction across subsequent time steps during occlusion periods. OC-SORT bridges the gap between the design philosophy of the Kalman filter and the constraints of real-world conditions, which is of great significance to the industry. It makes the Kalman filter more accurate while maintaining its elegant design and low computational overhead. Figure 5 illustrates the comprehensive framework of OC-SORT (Cao et al. 2023).

As shown in Fig. 5, in the data association process, Observation-Centric Momentum (OCM) refers to the incorporation of the directional consistency of the trajectory in the cost matrix. This inclusion aims to augment the velocity consistency cost, leading to improved matching of trajectories and observations. OCM is motivated by the design principle that estimating the direction between two points with extensive time intervals can mitigate sensitivity to noise. Additionally, Observation-Centric Recovery (OCR) represents a secondary association attempt between the last observation of an unmatched trajectory and the unmatched observation after the regular data association phase. For targets lost due to occlusion within a short period, OCR facilitates the recovery of the lost target regarding the observation from the frame before the loss. Observation-centric Re-Update (ORU) signifies that once a trajectory has been lost for a specific period and is then associated with an observation again, the time of its loss is retrospectively checked, and the Kalman Filter parameters are re-updated.

Matching strategies in multi-object tracking can be broadly categorized into one that seeks the optimal solution for a single allocation problem and another that considers multiple associations comprehensively. On the one hand, the approach aiming for the optimal solution in a single allocation problem focuses on finding the best solution among all potential associations without exploring every possibility. However, this method discards all information except the optimum, akin to the approach of graph neural network (GNN) utilizing greedy algorithms (He et al. 2021a; Scarselli et al. 2008). On the other hand, considering multiple association methods collectively aims to leverage more information by employing various association techniques and amalgamating them through a joint probability approach. Two traditional methods include the Joint Probability Data Association (Musicki and Evans 2004) and Multiple Hypothesis Tracking (Reid 1979).

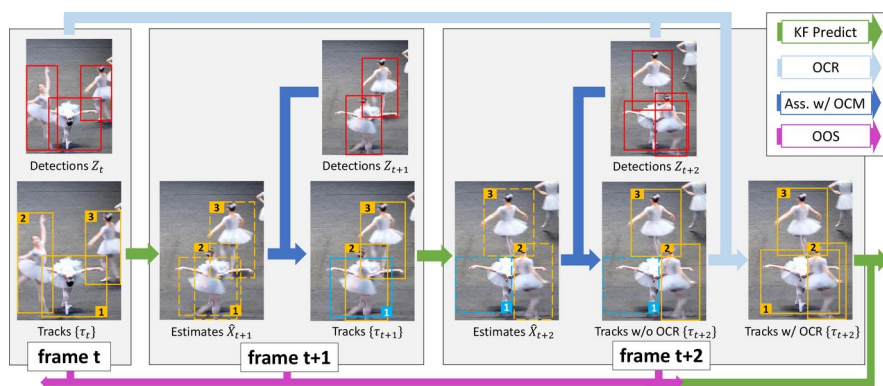


Fig. 5 Framework of OC-SORT (Cao et al. 2023)

Classical algorithms such as the Hungarian Algorithm (Henriques et al. 2011) and Nearest Neighbor Algorithm (Yu et al. 2016) are also prevalent. The Nearest Neighbor Algorithm clusters the closest set of values by establishing a threshold based on distance metrics (Takala and Pietikainen 2007; Fouzia et al. 2019) to match the target in the current frame with the tracked target in the previous frame. However, its performance diminishes in dense environments where distances between frames become comparable, leading to association errors. In contrast, the Hungarian Algorithm aims for the global optimum and is a combined optimization algorithm that efficiently solves allocation problems in polynomial time. It transforms the multi-object tracking data association issue into a bipartite graph minimum-weight matching problem, providing an optimal solution. Specifically, the Hungarian algorithm operates on a two-dimensional cost matrix where each element denotes the cost or price of assigning a task to a resource. It endeavors to discover an allocation that minimizes the overall cost by continuously searching for augmenting paths.

In 2021, Wang et al. (2021a) proposed LGMTracker, employing a reconstruct-to-embed strategy based on Graph Convolutional Networks to address the association problem for long-term tracking in the absence of appearance information, relying solely on motion information. The research also explores the significance of motion patterns for vehicle tracking. In 2022, Zhang et al. (2022c) proposed MTracker, utilizing position, motion, and appearance cues to compute similarity. Additionally, a more stable distribution is achieved by computing marginal probabilities based on appearance features, serving as a normalized distance that is more stable than the original feature distance. He et al. (2021a) proposed GMTracker, represents the association relationship as a general graph, and formulates the association problem as general graph matching. Specifically, the trajectory and intra-frame detection relationship is modeled as a general undirected graph. The association problem is then transformed into a general graph matching problem between the trajectory and detection graphs.

In 2023, Qin et al. (2023) proposed MotionTrack, which features an interaction module that learns interactive perceptual motion from short-term trajectories. A Refind module is also proposed to learn reliable long-term motion from the historical trajectories of the target, capable of associating interrupted trajectories with their corresponding detections, enabling the model to provide a reliable matching strategy even in extreme occlusion scenarios. In terms of trajectory matching, most methods strive to match the targets and trajectories within the current frame, often neglecting those trajectories that have lost targets due to long-term occlusions. It's not that they don't want to recover lost trajectories, but it's difficult to distinguish between lost trajectories and entirely new ones. Some methods use appearance features to re-associate lost targets with trajectories, but this introduces a significant amount of computation. MotionTrack provides a valuable approach by relying on the modeling of historical trajectories and complex motion to recover lost trajectories. Seidenschwarz et al. (2023) proposed GHOST, which combines ReID with a simple linear motion model using weighted sums, allowing each cue to receive more weight as needed for different datasets. This enables the standard ReID network to excel at appearance-based tracking. Cetintas et al. (2023) proposed SUSHI, which utilizes a graph neural network to process all levels of the hierarchy, where the GNN learns to leverage association cues related to the period and combine them. This renders the model uniform and highly generalizable across time scales, facilitating effective long-term tracking. Stadler and Beyerer (2023) proposed ImprAsso, proposed a Combined Matching approach that concurrently

considers all potential assignments in a single matching phase, and a novel Combined Distance metric is introduced to enhance the accuracy of the association, incorporating both motion and appearance information.

In 2024, Huang et al. (2024) proposed DeconfuseTrack, which integrates occlusion-aware non-maximum suppression to preserve a greater number of detections within occluded regions. This technique enriches the acquisition of motion and appearance information, thereby enhancing the probability of successful trajectory association and indirectly reducing the disarray stemming from missed detections. Liu et al. (2024a) proposed an online multi-object tracking model called FishTrack, which encodes the historical information of target positions using an encoder. It automatically updates the spatiotemporal information of the targets through an autoregressive mechanism, achieving effective integration of spatiotemporal data. Subsequently, a parallel dual-decoder architecture is employed to decode the motion and appearance features of the targets, mitigating the impact of occlusions and deformations. To address the challenge of multi-object tracking in maritime environments, Shao et al. (2024) propose a module inspired by motion compensation techniques used in airborne cameras. This module utilizes the intersection of dynamic spatial information of the ocean as a similarity metric. Additionally, a progressive refinement cascade matching strategy was designed to enhance the tracker's capability in handling multi-object tracking on the sea surface.

In addition, Lv et al. (2024) proposed a Decoupled Diffusion Motion Predictor (D2MP), which models the various motion distributions presented by the data as a whole. This method predicts the motion of a single object based on its historical motion information, thereby addressing complex non-linear motion. DiffMOT is the first method to introduce a diffusion probability model in multi-object tracking to handle non-linear motion prediction. Yang et al. (2024) proposed Hybrid-SORT, which compensates for strong cues by incorporating weak cues, effectively addressing the issue of spatial and appearance information becoming unclear due to high overlap between objects. This method introduces confidence and height states as potential weak cues, demonstrating strong generalizability across different trackers and scenes.

Table 1 provides an overview of the studies associated with multi-object tracking in the context of Tracking by Detection. Table 2 showcases the assessment outcomes of the multi-object tracking approach in Tracking by Detection, particularly on the MOT17 and MOT20 private test sets.

As illustrated in Table 1, we have compiled summaries of TBD MOT papers presented at ICCV, ECCV, and CVPR conferences over the past three years, covering topics such as detection and association. Additionally, we include code links to these papers. Furthermore, we offer a MOT toolkit for data and video transformation, visualization, and model performance evaluation. As shown in Table 2, on the test set of the MOT17 dataset, ImprAsso (Stadler and Beyerer 2023) achieved the highest MOTA metric at 82.2%, surpassing the second place by 1.1%. Similarly, SUSHI (Cetintas et al. 2023) attained the highest IDF1 metric at 83.1%, surpassing the second place by 1%. On the test set of the MOT20 dataset, ImprAsso (Stadler and Beyerer 2023) led with the highest MOTA metric at 78.6%, outperforming the second place by 0.6%. Additionally, SUSHI (Cetintas et al. 2023) achieved the highest IDF1 metric at 79.8%, exceeding the second place by 1%.

LGMTracker (Wang et al. 2021a), MTracker (Zhang et al. 2022c), ByteTrack (Zhang et al. 2022b), GHOST (Seidenschwarz et al. 2023), ImprAsso (Stadler and Beyerer 2023),

Table 1 Summary of tracking by detection MOT papers

Tracker	Publication	Detection	Association	Code Link ¹
LGMTracker (Wang et al. 2021a)	ICCV2021	CenterNet	Motion, Appearance	A-1
MTracker(Zhang et al. 2022c)	ECCV2022	Scaled-YO-LOv4	Location, Motion, Appearance	–
LPC-MOT (Dai et al. 2021)	CVPR2022	ResNet50-IBN	Graph Matching	A-2
GMTracker (He et al. 2021a)	CVPR2022	CenterNet	Graph Matching	A-3
ByteTrack (Zhang et al. 2022b)	ECCV2022	YOLOX	Location, Motion, Appearance	A-4
OC-SORT (Cao et al. 2023)	CVPR2023	YOLOX	Motion	A-5
MotionTrack (Qin et al. 2023)	CVPR2023	YOLOX	Location, Motion	–
GHOST (Seidenschwarz et al. 2023)	CVPR2023	YOLOX	Motion, Appearance	A-6
SUSHI (Cetintas et al. 2023)	CVPR2023	ResNet50-IBN	Graph Matching	A-7
ImprAsso (Stadler and Beyerer 2023)	CVPR2023	YOLOX	Motion, Appearance	–
DeconfuseTrack (Huang et al. 2024)	CVPR2024	YOLOX	Motion, Appearance	–
DiffMOT (Lv et al. 2024)	CVPR2024	YOLOX	Motion, Appearance	A-8
Hybrid-SORT (Yang et al. 2024)	AAAI2024	YOLOX	Location, Motion, Appearance	A-9

¹Appendix A contains the download link for Tracking by Detection MOT methods

DeconfuseTrack (Huang et al. 2024) and DiffMOT (Lv et al. 2024) utilize both motion and appearance information as data association cues. MTracker (Zhang et al. 2022c), ByteTrack (Zhang et al. 2022b) and Hybrid-SORT (Yang et al. 2024) additionally integrate location information, whereas motionTrack (Qin et al. 2023) utilizes both location and motion information. OC-SORT (Cao et al. 2023) relies exclusively on motion information for data association. Additionally, LPC-MOT (Dai et al. 2021), GMTracker (He et al. 2021a), and SUSHI (Cetintas et al. 2023) leverage Graph Matching to accomplish data association.

In TBD methods, feature extraction strategies are mainly divided into motion model-based feature extraction, appearance-based feature extraction, and joint detection and feature extraction. Motion model-based feature extraction is to estimate the position of the target in the next frame by motion prediction, using the intersection ratio or displacement distance of the detection frames of the neighboring frames as the matching basis. It only relies on geometric information and does not require a complex feature extraction network, and is suitable for the case of linear motion of the target in high frame rate videos. Appearance-based feature extraction utilizes CNNs to extract the target's apparent features and

Table 2 Results on MOT17 and MOT20 test sets: The table is left and right halves correspond to MOT17 and MOT20 private detection, respectively

The best result among published methods is in Bold
The second result among published methods is in Bold Italic

Tracker	MOT17		MOT20	
	MOTA	IDF1	MOTA	IDF1
MTracker (Zhang et al. 2022c)	77.3	75.9	66.3	67.7
GMTracker (He et al. 2021a)	66.2	70.6	65.0	68.7
ByteTrack (Zhang et al. 2022b)	80.3	77.3	77.8	75.2
OC-SORT (Cao et al. 2023)	78.0	77.5	75.5	75.9
MotionTrack (Qin et al. 2023)	81.1	80.1	78.0	76.5
GHOST (Seidenschwarz et al. 2023)	78.7	77.1	73.7	75.2
SUSHI (Cetintas et al. 2023)	81.1	83.1	74.3	79.8
ImprAsso (Stadler and Beyerer 2023)	82.2	82.1	78.6	78.8
DeconfuseTrack (Huang et al. 2024)	80.4	80.6	78.1	77.6
DiffMOT (Lv et al. 2024)	79.8	79.3	–	–
Hybrid-SORT (Yang et al. 2024)	79.9	78.7	76.7	78.4

performs cross-frame matching by cosine distance or Euclidean distance. Its performance is better in similar target scenes, but the strategy relies on high-quality detection frames and is susceptible to lighting changes and background interference. Joint detection and feature extraction is to output detection frames and Re-ID features in parallel in the target detection network, sharing the backbone network to reduce computational redundancy. Its detection and feature extraction are done synchronously, which is suitable for real-time tracking.

Despite the success of these methods, they have some common problems such as the balance between robustness and model complexity, and the lack of flexibility in different complex scenes of the same video. Moreover, the segmentation of the detection and tracking tasks of TBD methods into different training processes leads to global inconsistency, which degrades the overall performance. As a result, research on TBD methods has shown a decreasing trend in recent years. In contrast, the prevailing trend in MOT has transitioned from separate detection and tracking components to a unified model, i.e., E2E MOT. The E2E approach aims to bridge the gap between detection and tracking to ensure the stability and robustness of the model. As a result, the E2E approach is gradually replacing TBD MOT in many studies.

3.2 End-to-end

3.2.1 Multi-object tracking based on CNN

The utilization of CNNs for advancing multi-object tracking represents a robust methodology within the domain of computer vision. CNNs are widely recognized for their proficiency in handling extensive datasets, adeptly discerning intricate features, and extracting latent information (Wang et al. 2018). These networks find broad applications in various domains, such as object detection, feature extraction, spatio-temporal feature representation, and, more specifically, multi-object tracking. Within the Object Detection and Feature Extraction, the process of multi-object tracking consistently commences with object detection. During this pivotal phase, pre-trained deep convolutional architectures (Wang et al. 2021c; Gan et al. 2019; Xu et al. 2020; Zhang et al. 2020) are employed to identify objects

within video frames. These convolutional neural network models possess robust capabilities in feature extraction, substantially augmenting object detection precision.

Object ReID (Lusardi et al. 2021), which occurs when objects disappear and reappear within video sequences, significantly benefits from CNNs. By feeding object images into CNNs, discriminative feature representations are generated, leveraging learned object appearance models encompassing attributes like color, texture, and other pertinent features. These representations aid in verifying if the reappeared object matches previous identifications. Especially in Cross-camera tracking scenarios. As a supplement to matching criteria, ReID enriches tracking strategies, enhances the robustness of tracking methods against occlusions, and provides the possibility of recovering lost tracks. Specific methodologies for object ReID employ techniques like Siamese networks (Dong and Shen 2018) or triplet networks (Yin et al. 2020) to discern similarities between objects.

Figure 3 illustrates the CNN-based architecture of ResNet (He et al. 2016) and LSTM (Shi et al. 2015). FairMOT (Zhang et al. 2021) employs ResNet34 and ResNet50 as backbones, while UTM (You et al. 2023) utilizes ResNet101 as its backbone architecture. On the other hand, TripletTrack (Marinello et al. 2022) and MHT-bLSTM (Kim et al. 2018) leverage different methods for handling object motion. TripletTrack incorporates LSTM modeling, while MHT-bLSTM adopts a Bilinear methodology.

Shuai et al. (2021) proposed SiamMOT, including two variants of the Siamese tracker, namely, an implicit motion model (IMM) and an explicit motion model (EMM). The IMM utilizes the MLP to implicitly estimate instance-level motion between two frames, while the EMM employs a CNN to detect matching instances. Figure 6 illustrates the comprehensive framework structure of SiamMOT.

As illustrated in Fig. 6, the SiamMOT (Shuai et al. 2021) inputs are two frames I^t and $I^{t+\delta}$ and a set of detected instances R^t . The detection network outputs a set of detected instances $R^{t+\delta}$, while the tracker propagates R^t to time $t + \delta$ to generate $\tilde{R}_i^{t+\delta}$. Specifically, the region features $f_{R_i}^t$ of frame t , and the initial search region features $f_{S_i}^{t+\delta}$ of frame $t + \delta$ are passed through the Siamese Tracker τ to compute the channel dimensional inter-correlation and obtain the similarity of the two feature maps. Moreover, further predict the pixel-level confidence and location to get the prediction result $\tilde{R}_i^{t+\delta}$. Finally, the prediction result $\tilde{R}_i^{t+\delta}$ is simply matched with the $R_i^{t+\delta}$ detected by the detector Faster R-CNN (Ren et al. 2015). Siamese networks are a common paradigm used in many tasks that require

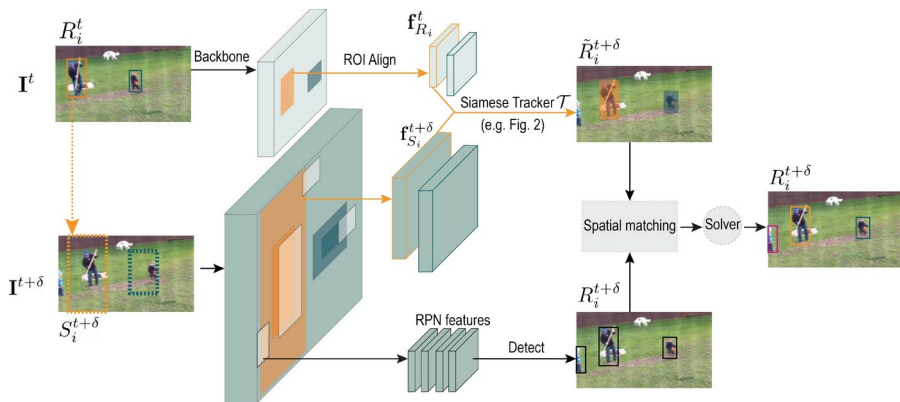


Fig. 6 Framework of SiamMOT (Shuai et al. 2021)

comparison or matching, and object tracking happens to fall into this category. Additionally, the structure of Siamese networks is relatively simple, with two branches of the network being very similar, resulting in comparable scale and parameter count, thus also possessing considerable inference speed.

Significant advancements in multi-object tracking based on CNNs have emerged in recent years. For instance, in 2023, You et al. (2023) proposed that UTM is an integrated tracking model that consolidates object detection, feature embedding, and identity association into a unified framework. The core of UTM lies in the Identity-Aware Feature Enhancement module, designed to augment detection and embedding processes by leveraging identity-aware insights. This module establishes a positive feedback loop, fostering mutual reinforcement among object detection, feature embedding, and identity association. By enhancing detection accuracy and refining feature embedding, the UTM ensures the generation of higher-quality tracking trajectories.

Following the rise of Transformer-based frameworks, a noteworthy reduction in recent research has been exclusively relying on CNNs for multi-object tracking, coinciding with the emergence of Transformer-based (Vaswani et al. 2017) frameworks. This trend could stem from inherent limitations within CNNs tailored for multi-object tracking. Initially, CNNs possess spatial constraints, limiting their effectiveness in processing image data within a confined spatial range. This spatial limitation results in a relatively constrained understanding of the global context around objects, presenting challenges in scenarios involving occlusions or substantial object movements. Objects situated further away may demand additional layers or a broader receptive field within CNNs, augmenting model complexity and necessitating a broader spectrum of data and computational resources.

In contrast, the Transformer architecture showcases distinct advantages in handling temporal data and adeptly manages material relationships among objects. Notably, the self-attention mechanism within the Visual Transformer dynamically integrates global contextual information and effectively discerns task-relevant data (Wang et al. 2022b). This capability empowers Transformers to navigate intricate relationships and establish long-range correlations within multi-object tracking scenarios. Consequently, the unique strength of the Transformer enhances efficiency in addressing complex tracking scenarios encompassing extended spatio-temporal dimensions. As a result, many multi-object tracking frameworks are adopting Transformer architectures or integrating both CNN and Transformer methodologies.

3.2.2 Multi-object tracking based on transformer

As the Transformer architecture (Vaswani et al. 2017) continues its advancement, particularly in natural language processing, its adaptation as the visual Transformer showcases outstanding performance across a wide array of visual tasks, including image classification (Chen et al. 2021), object detection (Carion et al. 2020), realistic image generation (Liu et al. 2022), audio-visual learning (Gan et al. 2020a, b), and various pixel-level assignments (Song et al. 2021). In multi-object tracking, an established methodology involves predicting the forthcoming location of objects based on preceding data, an area where the Transformer architecture demonstrates notable efficacy. Designed explicitly to handle sequential information, the Transformer adeptly executes frame-by-frame processing within the context of multi-object tracking tasks. Integrating the Transformer into multi-object tracking technol-

ogy (Kamboj 2024) requires the development of a specialized Tracker Architecture based on the Transformer model (Zhang et al. 2023b), which accommodates object detection (Li et al. 2022; Misra et al. 2021; Mao et al. 2021) and trajectory modeling (Shi et al. 2023).

In the realm of the object ReID (Tian et al. 2018; Liang et al. 2022) conventional CNNs often process local area features individually, potentially losing fine-grained information due to convolution and downsampling operations. In contrast, the Transformer method extracts features from each local area, resulting in a comprehensive global feature vector representing the entire image for subsequent ReID tasks (Wang et al. 2022a; He et al. 2021b; Ni et al. 2023). Additionally, the Transformer model facilitates trajectory and motion modeling by utilizing historical object state sequences as inputs. Leveraging the self-attention mechanism of the Transformer contributes to capturing spatial and temporal relationships between diverse objects (Zhu et al. 2022; Xu et al. 2019; Chu et al. 2023), enriching the understanding of object dynamics in multi-object tracking and enhancing the overall system performance.

Figure 3 demonstrates the classification of transformer-based MOT methods into two primary categories: Temporal and Graph methods. Specifically, Trackformer (Meinhardt et al. 2022) and TransTrack (Sun et al. 2020) belong to the category of MOT models utilizing Temporal mechanisms, whereas 3DMOTFormer (Ding et al. 2023) and ColTrack (Liu et al. 2023b) represent MOT models utilizing GNN. An influential work, Peize et al. (Sun et al. 2020) proposed TransTrack, presenting a Transformer-based framework tailored for joint detection and tracking. Utilizing the query-key mechanism of the Transformer, TransTrack efficiently generates object queries and tracking queries, thereby enhancing object association and refining tracking outcomes. The framework employs object and tracking queries to create a set of detection and tracking boxes, respectively. Utilizing a standard intersection over Union (IoU) matching strategy (Zou et al. 2022), commonly used in object detection and tracking tasks, TransTrack computes the intersection ratio between candidate bounding boxes generated by the object query and tracking query for each object. This process effectively determines the tracking box for objects. TransTrack framework is detailed in Fig. 7.

As shown in Fig. 7, TransTrack (Sun et al. 2020) establishes short tracks by utilizing two sets of detection results and box matching. It is logical to contemplate matching the tracking box with the object from the preceding frame to accentuate the temporal correlation in the tracking task. Two strategies are implemented to mitigate the impact of tracking box matching. One approach involves matching the initial tracking box to the object box in the previous frame via optimal bidirectional matching. The alternative strategy entails supervising the output of the tracking box with the current object box through optimal bidirectional

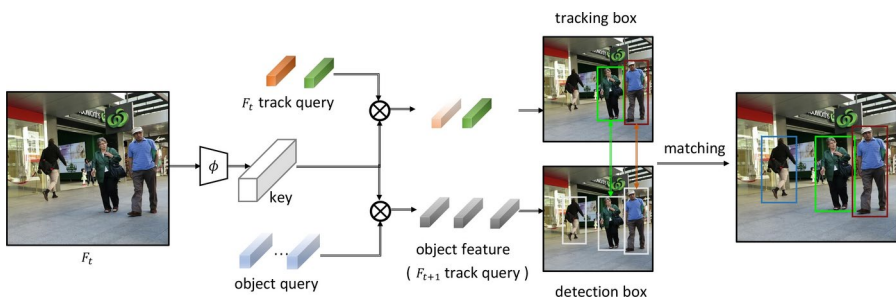


Fig. 7 Framework of TransTrack (Sun et al. 2020)

matching. The first strategy matches the initial tracking box with the object box from the previous frame through optimal bidirectional matching, which leverages historical information to predict the location of objects in the current frame, enhancing robustness against occlusions and short-term disappearances. The second strategy supervises the output of the tracking box with the current object box through optimal bidirectional matching, which helps in effective tracking and matching when there are significant changes in the target features. The combination of these two strategies allows TransTrack to improve the accuracy of target tracking in complex scenes while maintaining real-time performance.

The Transformer (Vaswani et al. 2017) architecture comprises encoders and decoders. The encoder focuses on capturing self-attention, denoted as query, key, and value in the Transformer model. Decoders, conversely, emphasize cross-attention, known as encoder-decoder attention. These two attention mechanisms effectively retain contextual information over long distances (Pu et al. 2022). Specifically, the decoder initializes new objects using static object queries, tracks existing trajectories and updates locations from tracking queries. Both query types utilize self-attention and cross-attention, incorporating global frame features. There are several reasons why Transformer now is popular in multi-object tracking, its self-attention mechanism enables the method capture long-range dependencies between objects in video sequences, which is crucial for tracking fast-moving or mutually occluded targets. Also, Transformer supports end-to-end training, eliminating the need for manual features and complex rules, and directly learns the tracking model from data, simplifying the training process and improving accuracy. In addition, Transformer encoder can extract rich target features, including appearance, motion, and relationships, thus generating robust object representations. Meinhardt et al. (2022) proposed TrackFormer, which encodes frame features with self-attention in the Transformer encoder.

Meanwhile, the query is decoded in the Transformer decoder using self-attention and cross-attention. The decoder alternates between self-attention for all queries, enabling joint reasoning about objects in the scene, and cross-attention, providing the query full access to the visual information of the encoded features. The advantage of self-attention mechanism lies in modeling global dependencies and enhancing features, making it suitable for target differentiation in dense scenarios. The advantage of cross-attention mechanism lies in aligning heterogeneous sequences and dynamic matching, making it suitable for detection-to-track association and cross-modal fusion.

Zeng et al. (2022) proposed MOTR, updating the trajectory query with self-attention and cross-attention for frame features. The updated trajectory query is further used to predict the bounding box, and the trajectory of an object is obtained from all prediction results of a trajectory query in different frames. In contrast, Zhou et al. (2022) proposed GTR, which eliminates the self-attention and cross-attention layers from the model, performing dot-multiplication of object features directly after the linear layer. This modification significantly reduces the AssA metrics. Luo et al. (2024) proposed DiffusionTrack, a method that indirectly applies an attention mechanism through a spatiotemporal fusion module in the diffusion head. This allows the same pair of bounding boxes to exchange temporal information, ensuring that data association is completed across two consecutive frames. Given the region of interest features of the current block and the self-attention output queries, linear inference and batch matrix multiplication are performed to obtain object queries.

The Multi-Head Attention mechanism allows the model to simultaneously focus on different parts of the subspace by employing multiple self-attentive layers to process inputs

in parallel. This enhances the representation capability of the model, enabling it to capture various patterns and relationships from the input data. The advantage of multi-head attention mechanism lies in feature learning and noise robustness, making it suitable for complex interaction scenarios.

GTR (Zhou et al. 2022) utilizes multi-head attention, a single-layer encoder, and a single-layer decoder, resulting in a lightweight network structure. Gao and Wang (2023) proposed MeMOTR, emphasizing the crucial role of multi-head attention in learning more discriminative representations, especially in scenarios with multiple similar objects in the same frame. Liu et al. (2023b) proposed ColTrack, introducing temporal blocking decoders that prevent interactions between historical tracking queries for the same target by modifying the attention mask of the multi-head attention module in the decoder. Ren et al. (2023) proposed FineTrack, proposing the Multi-head Part Mask Generator, which utilizes multi-head attention to filter background noise and extract global–local embeddings for a comprehensive target representation. A multi-branch parallel structure is implemented to efficiently and comprehensively focus on different target parts, obtaining diverse fine-grained representations.

Deformable attention (Xia et al. 2022), a novel mechanism adapting to different data structures by dynamically adjusting position weights to capture crucial information in the sequence, demonstrates superior performance, especially when dealing with complex data structures. The advantage of deformable attention mechanism lies in local focusing and computational efficiency, making it suitable for tracking fast-moving targets and real-time systems.

Xu et al. (2022) proposed TransCenter, featuring a decoder comprising Tracking Deformable Cross-Attention (TDCA) and Detection Deformable Cross-Attention (DDCA) modules. TDCA computes cross-attention between tracking queries and memories, generating tracking features. Similarly, DDCA computes cross-attention between detection queries and memories, producing detection features. Yu et al. (2022) proposed RelationTrack, introducing a Guided Transformer Encoder module that combines the powerful inference capability of the Transformer encoder and deformable attention to leverage global semantic relational information for associating detected targets. Wu et al. (2023) proposed TransRMOT, incorporating an early-fusion module in the encoder to densely integrate visual and linguistic features. A deformable attention layer follows this to refine further cross-modal representations for accommodating cross-modal inputs. Zhang et al. (2023a) proposed MOTRv2, building on the deformable DETR architecture (Zhu et al. 2020) and introducing trajectory and object queries. Object queries detect newborn or missing objects, while each trajectory query tracks unique instances over time.

Table 3 illustrates an overview of the studies associated with MOT in the context of End-to-End. Table 4 showcases the assessment outcomes of the MOT approach End-to-End, particularly on the MOT17 and MOT20 private test sets.

As shown in Table 3, over the past three years, we have compiled summaries of End-to-End MOT papers presented at ICCV, ECCV, and CVPR conferences, covering topics such as detection and association. Additionally, we include code links to these papers. As illustrated in Table 4, on the test set of the MOT17 dataset, UTM (You et al. 2023) achieved the highest MOTA metric at 81.8%, surpassing the second place by 1.8%. FineTrack (Ren et al. 2023) attained the highest IDF1 metric at 79.5%, surpassing the second place by 0.8%. On the test set of the MOT20 dataset, UTM (You et al. 2023) led with the highest MOTA metric

Table 3 Summary of end-to-end MOT papers

Tracker	Publication	Backbone	Attention	Code Link ¹
SiamMOT (Shuai et al. 2021)	CVPR2021	DLA-34	–	B-1
Relationtrack (Yu et al. 2022)	TMM2022	DLA-34	Deformable attention	–
GTR (Zhou et al. 2022)	CVPR2022	DLA-34	Multi-head attention	B-2
TrackFormer (Meinhardt et al. 2022)	CVPR2022	ResNet-50	Self-attention, cross attention	B-3
MOTR (Zeng et al. 2022)	ECCV2022	ResNet-50	Self-attention, cross-attention, multi-head attention	B-4
TransCenter (Xu et al. 2022)	TPAMI2022	PVTv2	Deformable attention	B-5
TransRMOT (Wu et al. 2023)	CVPR2023	ResNet-50	Self-attention, deformable attention	B-6
MOTRv2 (Zhang et al. 2023a)	CVPR2023	ResNet-50	Self-attention, deformable attention	B-7
MeMOTR (Gao and Wang 2023)	ICCV2023	ResNet-50	Self-attention, multi-head attention	B-8
UCSL (Meng et al. 2023)	ICCV2023	DLA-34	–	–
ColTrack (Liu et al. 2023b)	ICCV2023	ResNet-50	Multi-head attention	B-9
FineTrack (Ren et al. 2023)	CVPR2023	FAFPN	Multi-head attention	–
UTM (You et al. 2023)	CVPR2023	ResNet-101	–	–
Diffusion-Track (Luo et al. 2024)	AAAI2024	CSPDarknet	Self-attention, cross-attention	B-10

¹Appendix B contains the download link for End-to-End MOT methods

at 78.2%, outperforming the second place by 0.3%. FineTrack (Ren et al. 2023) achieved the highest IDF1 metric at 79.0%, exceeding the second place by 2.1%.

To assess the performance of state-of-the-art models in TBD MOT and E2E MOT, we meticulously select and analyze five models in each category. All six MOT benchmarks (Bernardin and Stiefelwagen 2008), (Luiten et al. 2021) can be obtained in Fig. 8 and specific values are listed in Tables 2 and 4. Note that for enhanced comparability, we normalize the metrics IDs and Frag-where lower values are preferable-by dividing them by 100. The resulting absolute values, IDs-T and Frag-T, are obtained by subtracting the normalized

Table 4 Results on MOT17 and MOT20 test sets: The table is left, and right halves correspond to MOT17 and MOT20 private detections, respectively

Tracker	MOT17		MOT20	
	MOTA	IDF1	MOTA	IDF1
SiamMOT (Shuai et al. 2021)	65.9	63.3	—	—
Relationtrack (Yu et al. 2022)	73.8	74.7	67.2	70.5
GTR (Zhou et al. 2022)	75.3	71.5	—	—
TrackFormer (Meinhardt et al. 2022)	74.1	68.0	—	—
MOTR (Zeng et al. 2022)	73.4	68.6	—	—
TransCenter (Xu et al. 2022)	76.2	65.5	72.9	57.7
MOTRv2 (Zhang et al. 2023a)	78.6	75.0	76.2	73.1
MeMOTR (Gao and Wang 2023)	72.8	71.5	—	—
UCSL (Meng et al. 2023)	73.0	70.4	62.4	63.0
ColTrack (Liu et al. 2023b)	78.8	73.9	—	—
FineTrack (Ren et al. 2023)	80.0	79.5	77.9	79.0
UTM (You et al. 2023)	81.8	78.7	78.2	76.9
DiffusionTrack (Luo et al. 2024)	77.9	73.8	72.8	66.3

The best result among published methods is in Bold

The second result among published methods is in Bold Italic



Fig. 8 The average performance comparison of the five Tracking by Detection MOT trackers and the five End-to-End MOT trackers under all the six authoritative MOT benchmarks (Bernardin and Stiefelhagen 2008; Luiten et al. 2021) to demonstrate their ability under various attributes. The specific values are in Tables 2 and 4. The two trackers in the outermost circle are ImprAsso (Stadler and Beyerer 2023) in TBD MOT and FineTrack (Ren et al. 2023) in E2E MOT, which have better performance in dealing with MOT challenges

values from 100. Additionally, we constrain FPS metrics within the 60–90 range, computing FPS-T as per Eq. 1.

$$FPS - T = \alpha \times FPS + \beta, \quad (1)$$

where α and β are used to adjust the proportion of different FPS, and we set $\alpha = 0.23$, $\beta = 56.32$ in Fig. 8a, set $\alpha = 1.00$, $\beta = 54.26$ in Fig. 8b.

As shown in Fig. 8, each tracker has its specialties in coping with different MOT challenges. Overall, ImprAsso (Stadler and Beyerer 2023) and FineTrack (Ren et al. 2023) are

the two trackers with the best real-time and overall solid performance. ImprAsso (Stadler and Beyerer 2023) and UTM (You et al. 2023) have the highest tracking accuracy. ByteTrack (Zhang et al. 2022b) and MOTRv2 (Zhang et al. 2023a) have the lowest Frag-T metrics because these two trackers have more track fragmentation during tracking.

The advantages of the Transformer architecture in E2E methods include global information relationship modeling, parallelism, and suitability for sequential tasks. First, Transformer, through its self-attention mechanism, captures dependencies between various positions in the input sequence, effectively modeling long-range dependencies. This characteristic allows Transformer to perform exceptionally well in tasks that require understanding of global context, such as machine translation and text summarization. Second, the self-attention layers in the Transformer model can be computed in parallel, making Transformer easier to implement for parallel processing to some extent, which could lead to faster processing in certain cases. Furthermore, due to its lack of sequential dependencies, Transformer can process all positions in the input sequence simultaneously, accelerating the training process. Lastly, although Transformer was originally designed to handle sequential data (e.g., natural language processing tasks), it can also be applied to certain image tasks, especially for image data with spatial relationships. This characteristic makes Transformer a highly versatile model architecture.

A surge in recent research on multi-object tracking employing E2E models. However, in the domain of multi-object tracking, object trajectories often extend across numerous temporal steps, necessitating the processing of longer sequences. The inherent limitation of Transformer models lies in their restriction on sequence length, potentially impeding their effectiveness in scenarios with prolonged sequences. Additionally, considerable computational resources are essential when managing many objects or processing high-resolution images. This challenge could prove significant, notably in embedded systems or applications requiring real-time responsiveness, where meeting stringent real-time demands might become difficult.

4 Datasets and evaluation metrics

4.1 Datasets

Research in multi-object tracking frequently necessitates extensive labeled datasets to assess and validate algorithmic performance. Ongoing advancements in data acquisition techniques have led to the introduction of diverse public datasets, aiming to propel the field of object tracking forward. Table 5 overviews commonly utilized public datasets tailored for multi-object tracking. These datasets encompass various scenarios and challenges, serving as crucial benchmarks to evaluate the performance and effectiveness of algorithms.

4.1.1 KITTI

The KITTI dataset (Geiger et al. 2013), acclaimed as the most significant autonomous driving scene dataset in the world, supports 3D tasks like object detection and 3D tracking. The dataset predominantly focuses on vehicular tracking, characterized by a video resolution of 1392×512 pixels, which facilitates high-definition analysis and recognition of vehicles

Table 5 Public datasets

Dataset	Year	Frames	Sequences	Tracklets	Link ¹	Type	Resolution
KITTI(Geiger et al. 2013)	2012	10870	50	977	C-1	Car,3D	1392×512
MOT15(Leal-Taixé et al. 2015)	2015	11283	22	1221	C-2	Pedestrians,2D	Multi
MOT16(Milan et al. 2016)	2016	11235	14	1276	C-3	Pedestrians,2D	Multi
MOT17(Milan et al. 2016)	2016	33705	14	1342	C-4	Pedestrians,2D	Multi
MOTS(Voigtlaender et al. 2019)	2019	5906	8	556	C-5	Generic,2D	Multi
TAO(Dave et al. 2020)	2020	4447038	3	16104	C-6	Generic,2D	1280×720
MOT20(Dendorfer et al. 2020)	2020	13410	8	3833	C-7	Pedestrians,2D	Multi
3D-ZeF(Pedersen et al. 2020)	2020	14398	8	32	C-8	Fish,3D	2704×1520
CroHD(Sundararaman et al. 2021)	2021	11464	9	5230	C-9	Pedestrians,2D	Multi
DanceTrack(Sun et al. 2022)	2022	105000	100	990	C-10	Pedestrians,2D	Multi
SN-Tracking(Cioppa et al. 2022)	2022	225375	201	5009	C-11	Soccer,2D	Multi
VOTS2023/24(Kristan et al. 2023)	2023	288000	144	—	C-12	Generic,2D	Multi

¹The download link for multi-object tracking datasets is in Appendix C

within the visual field. This extensive dataset is collected using a vehicle with a color/gray stereo camera, a 3D laser scanner, and diverse multi-sensors across various urban settings such as middle rings, suburbs, and highways. KITTI (Geiger et al. 2013) provides lidar point clouds and 3D bounding box trajectory points for multi-object tracking endeavors.

4.1.2 VOTS2023/24

In 2023, the Visual Object Tracking (VOT) project launched the Multi-Object Tracking Challenge (Kristan et al. 2016). However, VOTS2023 (Kristan et al. 2023) no longer distinguishes between single-object and multi-object tracking, nor between short-term and long-term tracking. Instead, it introduces a unified challenge that requires participants to perform segmentation tracking of one or more targets in long or short sequences, with targets potentially disappearing and reappearing later in the video. Additionally, the competition utilizes a new multi-object segmentation tracking dataset, providing participants with more challenging and diverse data to test the performance of trackers across different scenarios. In the competition, the VOTS2023/24 dataset consists of 144 sequences and a total of 341 targets. The average length of the sequence is about 2000 frames (minimum=63, maximum=10700, median=1810). Out of 144 sequences, 93 contain targets that have left the field of view at least once and then returned. Out of 341 targets, 168 experienced this situation.

4.1.3 MOT challenge

The MOTChallenge dataset (Dendorfer et al. 2021) is widely regarded as a prominent dataset in multi-object tracking, primarily focused on pedestrian multi-object tracking tasks. It comprises a suite of standard datasets, namely MOT15 (Leal-Taixé et al. 2015), MOT16 (Milan et al. 2016), MOT17 (Milan et al. 2016), and MOT20 (Dendorfer et al. 2020), facilitating evaluation in multi-object tracking. These datasets encompass scenarios involving pedestrians, and typical challenges like occlusions, crossings, and alterations in camera perspectives. The MOT15 (Leal-Taixé et al. 2015) dataset represents an enhanced iteration of its precursor, offering a spectrum of resolutions, including but not limited to 768×576 and 1224×370 , to accommodate diverse computational analyses. In contrast, the MOT16 (Milan et al. 2016) dataset introduces a novel corpus of footage, distinguished by denser pedestrian populations and an escalation in tracking complexity, with video resolutions standardized to 1920×1080 and 640×480 to facilitate advanced tracking methodologies. The MOT17 (Milan et al. 2016) dataset parallels MOT16 (Milan et al. 2016) in content yet expands the scope by integrating three additional detection models, aligning it with contemporary research paradigms in multi-object tracking. Furthermore, the MOT20 (Dendorfer et al. 2020) dataset delves into the intricacies of densely populated environments, establishing itself as a benchmark for datasets that demand sophisticated tracking solutions, with resolutions tailored to 1173×880 , 1920×734 , and analogous specifications.

In Fig. 9a, within the MOT17 dataset, sequences MOT17-06 and MOT17-14 exhibit low MOTA and HOTA metrics across all advanced models. Specifically, MOT17-06 and MOT17-14 pose additional challenges in the multi-camera data fusion process attributed to data acquisition from a mobile camera. Furthermore, in Fig. 10b, within the MOT20 dataset, MOT20-06 and MOT20-08 demonstrate low MOTA and HOTA metrics across advanced models. Notably, MOT20-06 and MOT20-08 impose substantial demands on MOT model



Fig. 9 a shows four sequences from the MOT17 test dataset, i.e., MOT17-03, MOT17-06, MOT17-08, and MOT17-14. b and c illustrate the results of six state-of-the-art methods in TBD and E2E MOT, respectively, evaluated on the MOT17 test dataset using MOTA and HOTA metrics

performance, stemming from many target pedestrians, overlapping appearance similarities among pedestrians, and considerable occlusion.

4.1.4 MOTS

The MOTS dataset (Voigtlaender et al. 2019) is an extension derived from annotating the MOT17 dataset. The dataset primarily focuses on tracking pedestrians, with video resolutions provided in 1920×1080 and 640×480 . It amalgamates object tracking with semantic segmentation, encompassing multiple moving objects across intricate scenarios that entail various challenges, including occlusion, rapid motion, and mutual obstruction among targets. This dataset predominantly concentrates on crowded scenes involving pedestrians, posing significant challenges owing to numerous instances of occlusion.

4.1.5 TAO

The TAO dataset (Dave et al. 2020) encompasses high-definition videos capturing 2907 distinct environments, averaging about half a minute in length, and spanning 833 diverse categories, a significant increase compared to existing datasets. The dataset features a broad spectrum of object categories for tracking, establishing itself as a versatile resource in the domain of general multi-object tracking. It utilizes high-definition video resolutions at 1920×1080 to enhance the precision of analytical processing. In multi-object tracking, categories typically derive from domains like autonomous driving and video surveillance, centering around specific object classes such as vehicles, pedestrians, and animals. However, these datasets often need to pay more attention to the presence of many real-world objects. In contrast, akin to the COCO dataset (Lin et al. 2014), the TAO dataset presents a wide range of categories, particularly in multi-object tracking, encompassing a broader and more real-world representation of object categories. Hence, the TAO dataset boasts considerable



Fig. 10 **a** shows four sequences from the MOT20 test dataset, i.e., MOT20-04, MOT20-06, MOT20-07, and MOT20-08. **b** and **c** illustrate the results of six state-of-the-art methods in TBD and E2E MOT, respectively, evaluated on the MOT20 test dataset using MOTA and HOTA metrics

scale and offers diversity in categories, presenting extensive resources and prospects for research and applications within multi-object tracking.

4.1.6 3D-ZeF

The 3D-ZeF dataset (Pedersen et al. 2020) is a stereo-based 3D RGB dataset tailored for multi-object zebrafish tracking studies within neurological disorder research. The dataset is specifically designed for tracking zebrafish, utilizing high-resolution video at 2704×1520 to enable sophisticated computational analysis and ensure detailed observation of subject movement. It comprises eight sequences varying in duration from 15 to 120 s, capturing the movement of 1 to 10 freely moving zebrafish. These video sequences are meticulously annotated with 86,400 data points and bounding boxes, delivering a comprehensive account and characterization of zebrafish movement trajectories.

4.1.7 CroHD

The CroHD dataset (Sundaraman et al. 2021) is tailored explicitly for head tracking within crowded scenes characterized by high crowd densities. This dataset is specifically oriented towards pedestrian tracking, videos including a variety of resolutions. It incorporates diverse indoor and outdoor settings under varying lighting and environmental conditions. Each sequence maintains a frame rate of 25fps. This extensive dataset comprises nine sequences encompassing 11,463 frames, capturing over 2,276,838 heads in distinct scenarios. These frames are meticulously annotated, offering 5,230 detailed tracks. The primary goal of the CroHD is to present a spectrum of crowded scenarios with high crowd densities. Researchers can utilize this dataset to explore and enhance algorithms dedicated to head tracking within complex environments.

4.1.8 DanceTrack

The DanceTrack dataset (Sun et al. 2022) is an extensive multi-object tracking dataset developed to tackle complex tracking scenarios involving individuals under challenging conditions like occlusion, frequent crossings, similar attire, and diverse body postures. This dataset is primarily dedicated to the tracking of pedestrians, encompassing video data captured at multiple resolutions to facilitate a comprehensive analytical approach in the domain of computer vision. Its primary focus is highlighting the significance of motion analysis within multi-object tracking. By offering challenging scenarios, this dataset aids researchers in comprehending and addressing the complexities inherent in real-world multi-object tracking applications, particularly regarding occlusion and crowd crossings.

4.1.9 SN-tracking

The SN-Tracking dataset (Cioppa et al. 2022) comprises 200 sequences, each lasting 30 s, covering 11 soccer action categories like direct free kicks, fouls, etc. Focused on the tracking of pedestrians and footballs, the dataset incorporates video sequences available in various resolutions. These sequences represent challenging tracking scenarios in soccer. The dataset includes a 45-minute halftime break, providing opportunities for long-term tracking

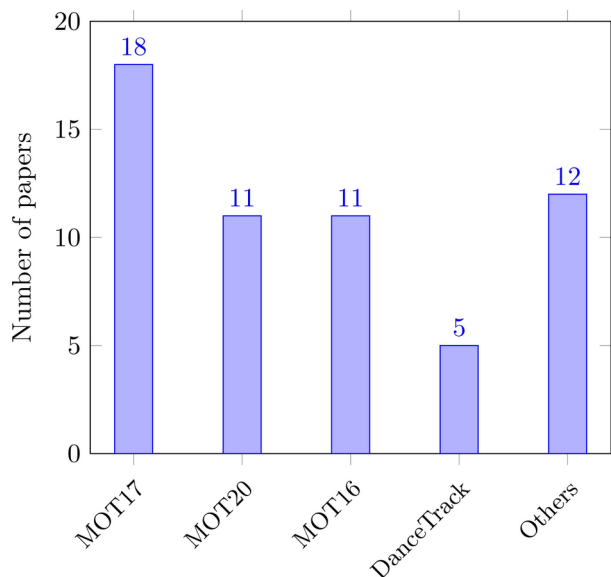
evaluation. Fully annotated with bounding boxes and track IDs at 25fps, the dataset facilitates training MOT baselines specific to soccer. It presents challenges such as multi-player, referee, and ball tracking under fast motion or heavy occlusion conditions.

The analysis portrayed in Fig. 11 reveals that the MOT17 dataset is notably predominant and more frequently employed than other datasets in research papers focusing on multi-object tracking. Interestingly, recent observations indicate a growing trend in utilizing the DanceTrack dataset introduced in 2022. Simultaneously, there is a discernible decrease in the frequency of usage of older datasets such as MOT16 and MOT15 in multi-object tracking research papers.

In the aforementioned common datasets, the usage frequency of the MOT16 and MOT15 datasets has decreased due to some shortcomings. For example, the distribution of crowd density in the training and testing sequences is imbalanced, and some methods overfit certain sequences (e.g., PETS09-S2L1), making them unsuitable for improving the capability of object tracking models to handle occlusions. On the other hand, the MOT17 dataset provides more crowded scenes, different viewpoints, camera movements, and varying weather conditions. The annotated objects not only include pedestrians but also vehicles, seated people, occluded objects, and other important object categories. This enables accurate calculation of the occlusion levels and cropping of all bounding boxes, thereby enhancing the performance of object tracking models in dealing with challenges such as occlusion, lighting changes, and camera movements.

The MOT20 dataset features highly crowded scenes with significant occlusion and target overlap, presenting tracking challenges that help improve the model's ability to handle extremely dense scenarios during training. The MOT20 test set includes sequences from both known and unknown scenes, which further aids in enhancing the model's generalization ability for evaluating the performance of both detectors and trackers during training. The DanceTrack dataset mainly consists of group dance videos, which exhibit frequent variations in motion speed, thus helping the model improve its ability to handle motion blur during training. This dataset also includes a significant amount of nonlinear motion, with

Fig. 11 Statistics on the number of times the Table 5 dataset was used in a total of 36 MOT papers from CVPR2021 to CVPR2023



frequent relative position changes, which assists the model in enhancing its tracking capabilities for nonlinear motion patterns during training.

Additionally, the MOT series datasets (e.g., MOT17, MOT20) are primarily collected from surveillance perspectives, covering public spaces such as city streets. These datasets are characterized by wide fields of view and fixed cameras, where the targets in the scenes typically exhibit a clear spatial layout. As such, they are highly applicable in intelligent security fields, where they can accurately track pedestrian movement trajectories and identify potential threats in urban environments, transportation hubs, and commercial areas. In contrast, the DanceTrack dataset is typically captured in scenarios involving nonlinear and variable-speed motion. This makes it highly relevant for applications in sports events, such as analyzing the movements of athletes like skiers. It provides valuable tactical decision support and performance analysis for coaches, athletes, and other analysts.

Currently, the mainstream researches are focused on MOT and VOT. Among them, MOTChallenge is the most widely used dataset in the MOT field, primarily targeting multi-object tracking for pedestrians. Since MOT17 provides only three detectors, it is considered relatively fair and is one of the most popular datasets in current research. On the other hand, VOTChallenge no longer distinguishes between single-object and multi-object tracking. It requires tracking algorithms to perform segmentation tracking of one or more general objects in both short and long sequences. Additionally, targets may disappear and reappear later in the video, which challenges the tracker's ability to maintain continuous tracking and segmentation of objects in complex scenarios.

4.2 Evaluation metrics

The *Multiple Object Tracking Precision (MOTP)* (Bernardin and Stiefelbogen 2008) functions as an evaluation metric for assessing the precision of multiple object tracking methodologies. It quantifies the score by measuring how accurately the tracker localizes the objects. Notably, this metric evaluates the precision of object positions regardless of the ability of trackers to perceive object arrangements or maintain a consistent trajectory. A higher *MOTP* value indicates a more precise estimation of object positions achieved by the tracking algorithm.

$$MOTP = \frac{\sum_{i,t} d_t^i}{\sum_t c_t} \quad (2)$$

In Eq. 2, the variable d represents the distance, indicating the dissimilarity between the detected object i coordinates in the t frame and the actual label information. Usually, this value is computed using strategies such as intersection ratio matching or Euclidean distance. The variable c denotes the count of matched objects in the t frame.

The *Multiple Object Tracking Accuracy (MOTA)* (Bernardin and Stiefelbogen 2008) is one of the most commonly used evaluation metrics in MOT, which provides a comprehensive assessment of a tracking algorithm's performance in tracking multiple objects. *MOTA* takes into account several factors, such as False Negatives (*FN*), False Positives (*FP*), and ID Switches (*IDS*), to quantify the impact of these tracking errors on the tracking task. A higher *MOTA* value indicates better performance of the tracking algorithm.

$$MOTA = 1 - \frac{\sum_t (FN_t + FP_t + IDSW_t)}{\sum_t GT_t} \quad (3)$$

In Eq. 3, where t represents the number of frames. *False Negative* (FN) represents the count of objects that the algorithm failed to correctly detect in the t frame, while *False Positive* (FP) denotes the count of objects that the algorithm has incorrectly seen in the t frame. The term *Identification switch* ($IDSW$) indicates the number of identifier switches occurring in the t th frame. GT_t is the number of ground truth objects. FN and FP primarily evaluating detector performance, while $IDSW$ reflects the joint impact of detector and tracker performance.

Identification Precision (IDP) is a metric that evaluates the accuracy of pedestrian identification within individual pedestrian frames. A higher IDP value indicates a more precise allocation of identification by the tracking algorithm. On the other hand, *Identification Recall* (IDR) quantifies the accuracy of pedestrian identification within individual pedestrian boxes. A higher IDR value indicates that the tracking algorithm is more capable of identifying and tracing the identity of the object.

Identification F1 Score ($IDF1$) (Luiten et al. 2021) assesses the correctness of detection and ground truth, considering both the accuracy of detection and tracking along with the missed detection rate. A higher $IDF1$ value indicates that the tracking algorithm is more effective in maintaining consistent object identification.

$$IDP = \frac{IDTP}{IDTP + IDFP} \quad (4)$$

$$IDR = \frac{IDTP}{IDTP + IDFN} \quad (5)$$

$$IDF1 = \frac{2IDTP}{2IDTP + IDFP + IDFN} \quad (6)$$

Identification True Positives ($IDTP$) in Eqs. 4, 5, 6 signify the count of frame pairs accurately assigned with the exact identification. When the algorithm correctly gives the same identification to an object across consecutive frames, these instances contribute to $IDTP$.

Identification False Positives ($IDFP$) arise when the tracking algorithm incorrectly associates an unknown object identifier. Conversely, *Identification False Negatives* ($IDFN$) occur when the tracking algorithm fails to match a known object identifier correctly, losing objects in specific frames.

Higher Order Tracking Accuracy ($HOTA$) (Luiten et al. 2021) is a newly proposed evaluation metric designed to address some of the shortcomings of $MOTA$, particularly in scenarios with frequent ID switches. $HOTA$ improves upon traditional metrics by decomposing the tracking task into three sub-tasks: detection, association, and localization. This decomposition allows for a more precise evaluation of the overall performance of multi-object tracking systems. In addition to considering traditional factors like False Negatives (FN), False Positives (FP) and ID Switches (IDS), $HOTA$ further accounts for trajectory quality and target matching accuracy, providing a more comprehensive and precise assessment of tracking algorithms.

$$HOTA = \int_0^1 HOTA_{\alpha} d\alpha \approx \frac{1}{19} \sum_{\alpha \in \{0.05, 0.1, \dots, 0.9, 0.95\}} HOTA_{\alpha} \quad (7)$$

In Eq. 7, $HOTA_{\alpha}$ refers to the *HOTA* metric computed for a specific value of α . It involves determining *HOTA* values across a range of α values (from 0 to 1) and averaging them. Further computations are performed to derive the final *HOTA* score.

$$HOTA_{\alpha} = \sqrt{\frac{\sum_{c \in \{TP\}} A(c)}{|TP| + |FN| + |FP|}} \quad (8)$$

$$A(c) = \frac{|TPA(c)|}{|TPA(c)| + |FNA(c)| + |FPA(c)|} \quad (9)$$

In Eq. 8, $A(c)$ represents the extent of similarity between predicted and actual trajectories. *True Positive (TP)* signifies the condition where the ground truth detection and the expected detection match if the localization similarity equals or exceeds a predefined threshold. Equation 9 outlines the specific components: *False Positive Association (FPA)* refers to the collection of ID false positives that misidentify a non-target entity as a target, *True Positive Association (TPA)* indicates the collection of ID true positives that accurately track the target and assign the correct identity, while *False Negative Association (FNA)* represents the collection of ID false negatives that fail to identify and track the target appropriately.

In terms of the calculation formulas, the *MOTA* metric places more emphasis on the overall tracking accuracy, especially the accuracy of object detection. On the other hand, the *HOTA* metric mainly focuses on the trajectory correlation and identity consistency. From the perspective of practical application scenarios, compared with the *HOTA* metric, the *MOTA* metric places more emphasis on the tracking accuracy of the model. Therefore, it is more suitable for fields such as autonomous driving and intelligent security monitoring, where tracking accuracy is of great concern. In contrast, the *HOTA* metric focuses more on the model's association ability over long distances. As a result, it is more applicable to the field of tactical analysis in sports events, where global aspects are highly valued.

5 Future directions

MOT has garnered considerable attention in recent years, fostering a substantial body of research. However, it remains a dynamic and evolving field with numerous uncharted prospects. This paper delineates promising trajectories in MOT, including multiple cameras, real-world, medical imaging and cross-modal directions.

5.1 MOT under multiple cameras

Multi-Object Tracking (MOT) in multi-camera environments holds significant practical importance (Cheng et al. 2023; Wang et al. 2023b; Nguyen et al. 2022, 2024). By integrating the fields of view from multiple cameras, a larger scene can be covered, ensuring continuous

monitoring of targets and eliminating blind spots. Multi-camera data fusion allows targets to be tracked from multiple perspectives, reducing errors caused by individual cameras or occlusion issues, thereby improving the robustness and accuracy of target tracking.

Apart from multi-object tracking within a single camera, multi-camera task of re-identification is also critical for cross-camera tracking. In this study, experiments were conducted on the classical multi-camera task of re-identification, as shown in Table 6:

In Table 6, re-identification methods integrated with wavelet convolution (Finder et al. 2025) demonstrated a certain degree of improvement in cross-camera re-identification performance. This enhancement is attributed to the low-frequency information provided by wavelet convolution, which helps strengthen the correlation between features and optimize retrieval results. Moreover, the Re-ranking algorithm (Zhong et al. 2017), which extends single-to-multiple re-identification tasks to many-to-many retrieval tasks, further improved the results of cross-camera re-identification tasks.

In summary, the incorporation of wavelet convolution and the Re-ranking algorithm proved beneficial for multi-object tracking. Therefore, combining re-identification with single-camera multi-object tracking can facilitate cross-camera multi-object tracking. In multi-object tracking tasks, trajectory fragments provide rich information, indicating that further exploration of low-frequency feature analysis and many-to-many retrieval methods remains a promising direction.

5.2 MOT with real-world

Multi-object tracking (MOT) with real-world refers to the task of simultaneously tracking multiple objects in complex and dynamic real-world environments using computer vision techniques. This differs from multi-object tracking in idealized laboratory settings, as real-world MOT must deal with more uncertainty and complexity.

In real-world scenarios, relying solely on two-dimensional (2D) information is far from sufficient due to its inherent limitations, including occlusion, viewpoint variations, and ambiguous spatial relationships. These issues are particularly pronounced in complex, real-world environments, where solely depending on 2D appearance features makes it difficult to maintain target consistency, often leading to missed detections, false positives, or ID switches. Consequently, incorporating three-dimensional (3D) information into multi-object tracking tasks becomes essential. To this end, this paper introduces depth information into the multi-object tracking task and conducts experiments. The results are shown in Table 7:

In Table 7, building on Carion et al. (2020), this paper introduces the Depth Information (DI) to enhance the feature map's information representation. Additionally, the Part Appearance Information (PAI) is used to enhance both the body appearance features and limb appearance features of the target, enriching the association constraints. The experiments demonstrate that incorporating depth information in MOT and improving appearance

Table 6 Experiments on Veri-776 (Liu et al. 2016b)

Method	mAP	Rank-1
UFDN(Qian et al. 2022)	80.1	95.9
UFDN(Qian et al. 2022) + WtConv	80.5	96.5
UFDN(Qian et al. 2022) + WtConv + Re-ranking	85.6	97.0
TransReID(He et al. 2021b)	80.2	97.1
TransReID(He et al. 2021) + WtConv	82.3	97.7
TransReID(He et al. 2021b) + WtConv + Re-ranking	91.2	98.0

feature learning lead to an increase in tracking accuracy. This indicates that different detection methods can influence tracking accuracy to some extent. How to better detect tracking targets or extend detection methods from the 2D to the 3D domain, thereby alleviating occlusion issues in multi-object tracking, is a critical problem that warrants further exploration. Additionally, how to better coordinate global and local information to obtain more distinctive appearance features is also an important approach to optimizing multi-object tracking performance.

Of course, the real-time performance of multi-object tracking tasks is equally crucial in real-world applications. Real-time multi-object tracking systems need to detect and track multiple targets quickly and accurately in dynamic environments. Attention mechanism based tracking methods, although excellent in feature attention, have high time complexity, resulting in slow inference. Transformer-based architectures like Trackformer(Meinhardt et al. 2022) are computationally huge and not suitable for real-time applications. Traditional detection-based tracking methods, such as ByteTrack(Zhang et al. 2022b), are computationally efficient, but the accuracy needs to be optimized, making it difficult to strike a balance between speed and accuracy.

In addition, due to challenges such as high computational complexity, target density and interaction, as well as non-linear and variable motion of objects, it is difficult to overcome issues like the complexity of dynamic real-world scenes and large volumes of data. Potential solutions include adopting lightweight network architectures to reduce model parameters and computational load. Lightweight networks (Li et al. 2023a; Liu et al. 2023a) are crucial in real-world applications with limited computational resources, and techniques such as pruning, quantization, and knowledge distillation can be employed to achieve this. Additionally, continuous learning (Xiao et al. 2024) that improves generalization ability can enhance the real-time performance of the model in complex real-world environments. In resource-constrained environments, balancing speed and accuracy continues to be a primary objective in MOT.

5.3 MOT in medical imaging

Multi-Object Tracking in Medical Imaging (Wan et al. 2021; Jiang et al. 2024; Feng et al. 2024) refers to the process of tracking multiple targets (such as organs, tumors, blood vessels, or other anatomical structures) in medical image sequences. This task is crucial for medical applications such as diagnosis, surgical navigation, and disease monitoring. Especially in time-series data (such as MRI, CT, and ultrasound images), accurately tracking different targets helps provide more precise medical analysis and decision-making.

In medical image processing, object tracking algorithms improve disease diagnosis efficiency and accuracy by real-time tracking of the spatial position and temporal changes of targets. They also optimize the treatment process, reduce manual intervention, and enhance

Table 7 Experiments on MOT17 (Milan et al. 2016)

DI	PAI	MOTA(%)	AssA(%)	IDF1(%)
×	×	67.4	50.5	63.4
✓	×	69.8	51.1	65.2
✓	✓	71.8	51.6	66.8

The DI is Depth Information (DI)

The PAI is part appearance information

the quality of healthcare. This paper explores the impact of tracking methods in the medical imaging domain, using ULM restoration as an example, as shown in Fig. 12:

In Fig. 12a Liu and Almekkawy (2023) is used as the baseline, and (b) the method from Wojke et al. (2017) is added for comparison. Due to the high similarity between microbubbles, (c) fine-tuning was performed on our simulation dataset to further optimize the model's parameters and weights. It can be observed that the images are restored with more detailed and complex vascular structures.

In medical image restoration, the integration of multi-object tracking with super-resolution localization microscopy enables precise localization and dynamic tracking at the cellular or subcellular level. The application of multi-object tracking technology in medical image restoration, especially in super-resolution localization microscopy, holds significant potential for development. By combining advanced deep learning methods, temporal analysis, multimodal data fusion, and other technologies, it provides strong support for precision medicine, personalized treatment, and early disease diagnosis.

5.4 MOT with cross-modal

Compared to traditional unimodal models (such as CNNs and Transformers) that only extract visual features from images, which are limited to using visual features for MOT without fully understanding and processing the information, cross-modal models (Wang et al. 2023a) operate by integrating different types of input (e.g., text and other data) simultaneously. These models use CNNs and Transformers to extract textual features, leveraging linguistic cues to fully capture the complexity of context.

Currently, cross-modal models have been widely applied in tasks such as visual grounding (Yang et al. 2022) and single-object tracking (Li et al. 2023c). The comparison of classic cross-modal visual grounding and cross-modal single-object tracking methods is shown in Table 8:

In Table 8, by comparing these representative works in the table, it is evident that they all adopt cross-modal fusion structures. VLT_{TT} (Guo et al. 2022) proposed the Modal Mixer and Asymmetric Search Strategy. TransVLT (Zhao et al. 2023a) utilized a Transformer-based network with proxy tokens to guide the cross-modal fusion module. JointNLT (Zhou

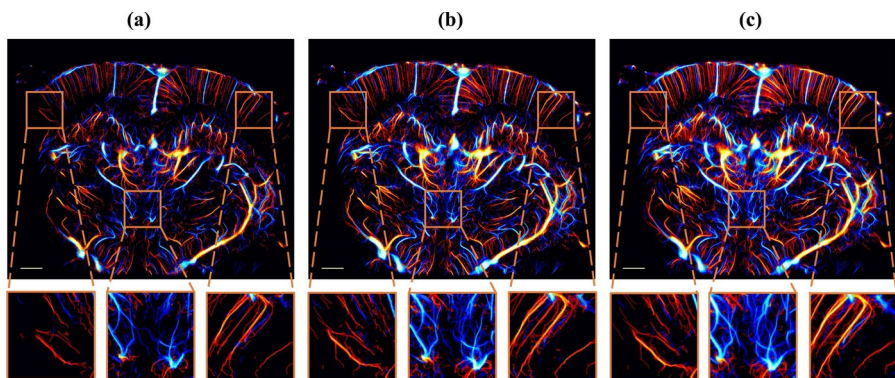


Fig. 12 Visualization in the OPULM-PALA (Heiles et al. 2022). **a** is the visualization of the baseline (Liu and Almekkawy 2023). **b** is the visualization of the method which replaces the tracking part in (a) with (Wojke et al. 2017). **c** is the visualization of method which makes parameter adjustments based on (b)

Table 8 Methods with cross-modal on TNL2K(Wang et al. 2021b)

Method	AUC	P_{Norm}	P
VLT _{TT} (Guo et al. 2022)	54.7	71.8	55.3
TransVLT(Zhao et al. 2023a)	56.0	61.7	—
JointNLT(Zhou et al. 2023)	56.9	73.6	58.1
MMTrack(Zheng et al. 2023)	58.6	75.2	59.4

et al. 2023) employed a multi-source relationship modeling module and a semantic-guided temporal modeling module. MMTrack(Zheng et al. 2023) reformulated the visual-linguistic tracking task as a token generation task. However, these models may struggle to accurately locate targets when dealing with ambiguous natural language descriptions. Furthermore, they still have limitations in understanding the deep relationships between language semantics and visual features, failing to capture the subtle semantic differences in language descriptions that affect object tracking.

The challenge in cross-modal learning lies in the difficulty of performing meaningful feature alignment and high-correlation feature fusion across different feature spaces. A review of existing works reveals that current alignment methods mostly rely on contrastive learning to impose data distribution constraints (Ma et al. 2024), while often neglecting a deeper understanding of the task itself. To address these issues, semantic-level feature alignment that enhances foreground-background relationship understanding, or relationship-guided feature fusion, are both promising avenues for further exploration. He et al. (2024) proposed DeepRMOT, which incorporates deep cross-modal fusion to address the limitations of previous methods that performed cross-modal fusion at only a single stage. By employing a bidirectional cross-modal encoder and a cross-modal decoder, the model strengthens visual-language representations, thereby enhancing object detection and data association capabilities.

In addition, Li et al. (2024) proposed Semantic Multi-Object Tracking (SMOT), which aims to estimate object trajectories while capturing the semantic details of related trajectories. The BenSMOT dataset was introduced as the first publicly available benchmark for SMOT, containing 3,292 videos and 151K frames, covering a wide range of scenes for human semantic tracking. For each video sequence, BenSMOT provides annotations of object trajectories, along with corresponding natural language instance captions, instance interactions, and overall titles.

6 Conclusion and discussion

This manuscript presents a concise yet comprehensive overview of recent advancements in computer vision, specifically focusing on multi-object tracking. The review provides an in-depth analysis of critical challenges, addressing concerns such as occlusion handling and variations in lighting conditions. It thoroughly explains the fundamental principles of convolutional neural network-based and Transformer-based multi-object tracking, meticulously outlining associated datasets and evaluation metrics. Additionally, it draws insights from emerging trends in multi-object tracking to uncover limitations in existing methodologies and explore potential paths for future investigation. Contemporary research is increasingly inclined towards cross-modal architectures due to their information-rich nature and robust contextual awareness, significantly enhancing the stability and performance of tracking sys-

tems. However, recent studies have highlighted challenges related to resource allocation, urging a focused approach within lightweight architectures. In summary, this manuscript serves as a compendium for researchers delving into the intricate domain of multi-object tracking. It aims to assist those contributing to this dynamic and expanding field by consolidating recent advancements and highlighting promising trajectories.

Appendix A: The download link for tracking by detection MOT methods

1. LGMTracker (Wang et al. 2021a): <https://github.com/GaoangW/LGMTracker>
2. LPC-MOT (Dai et al. 2021): https://github.com/daip13/LPC_MOT
3. GMTracker (He et al. 2021a): <https://github.com/jiaweihe1996/GMTracker>
4. ByteTrack (Zhang et al. 2022b): <https://github.com/ifzhang/ByteTrack>
5. OC-SORT (Cao et al. 2023): https://github.com/noahcao/OC_SORT
6. GHOST (Seidenschwarz et al. 2023): <https://github.com/dvl-tum/GHOST>
7. SUSHI (Cetintas et al. 2023): <https://github.com/dvl-tum/SUSHI>
8. DiffMOT (Lv et al. 2024): <https://github.com/Kroery/DiffMOT>
9. Hybrid-SORT (Yang et al. 2024): <https://github.com/ymzis69/HybridSORT>

Appendix B: The download link for end-to-end MOT methods

1. SiamMOT (Shuai et al. 2021): <https://github.com/amazon-science/siam-mot>
2. GTR (Zhou et al. 2022): <https://github.com/xingyizhou/GTR>
3. TrackFormer (Meinhardt et al. 2022): <https://github.com/timmeinhardt/trackformer>
4. MOTR (Zeng et al. 2022): <https://github.com/megvii-research/MOTR>
5. TransCenter (Xu et al. 2022): <https://github.com/yihongXU/TransCenter>
6. TransRMOT (Wu et al. 2023): <https://github.com/wudongming97/RMOT>
7. MOTRv2 (Zhang et al. 2023a): <https://github.com/megvii-research/MOTRv2>
8. MeMOTR (Gao and Wang 2023): <https://github.com/MCG-NJU/MeMOTR>
9. ColTrack (Liu et al. 2023b): <https://github.com/bytedance/ColTrack>
10. DiffusionTrack (Luo et al. 2024) <https://github.com/RainBowLuoCS/DiffusionTrack>

Appendix C: The download link for multi-object tracking datasets

1. KITTI (Geiger et al. 2013): https://www.cvlibs.net/datasets/kitti/eval_tracking.php
2. MOT15 (Leal-Taixé et al. 2015): <https://motchallenge.net/data/MOT15/>
3. MOT16 (Milan et al. 2016): <https://motchallenge.net/data/MOT16/>
4. MOT17 (Milan et al. 2016): <https://motchallenge.net/data/MOT17/>
5. MOTS (Voigtlaender et al. 2019): <https://motchallenge.net/data/MOTS/>
6. TAO (Dave et al. 2020): https://motchallenge.net/data/TAO_Challenge/
7. MOT20 (Dendorfer et al. 2020): <https://motchallenge.net/data/MOT20/>
8. 3D-ZeF (Pedersen et al. 2020): <https://motchallenge.net/data/3D-ZeF20>
9. CroHD (Sundararaman et al. 2021): https://motchallenge.net/data/Head_Tracking_21/

10. DanceTrack (Sun et al. 2022): <https://github.com/DanceTrack/DanceTrack>
11. SoccerNet (Cioppa et al. 2022): <https://www.soccer-net.org/data>
12. VOTS2023/24 (Kristan et al. 2016): <https://www.votchallenge.net/vots2023/dataset.htmlhttps://www.votchallenge.net/vots2024/dataset.html>

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